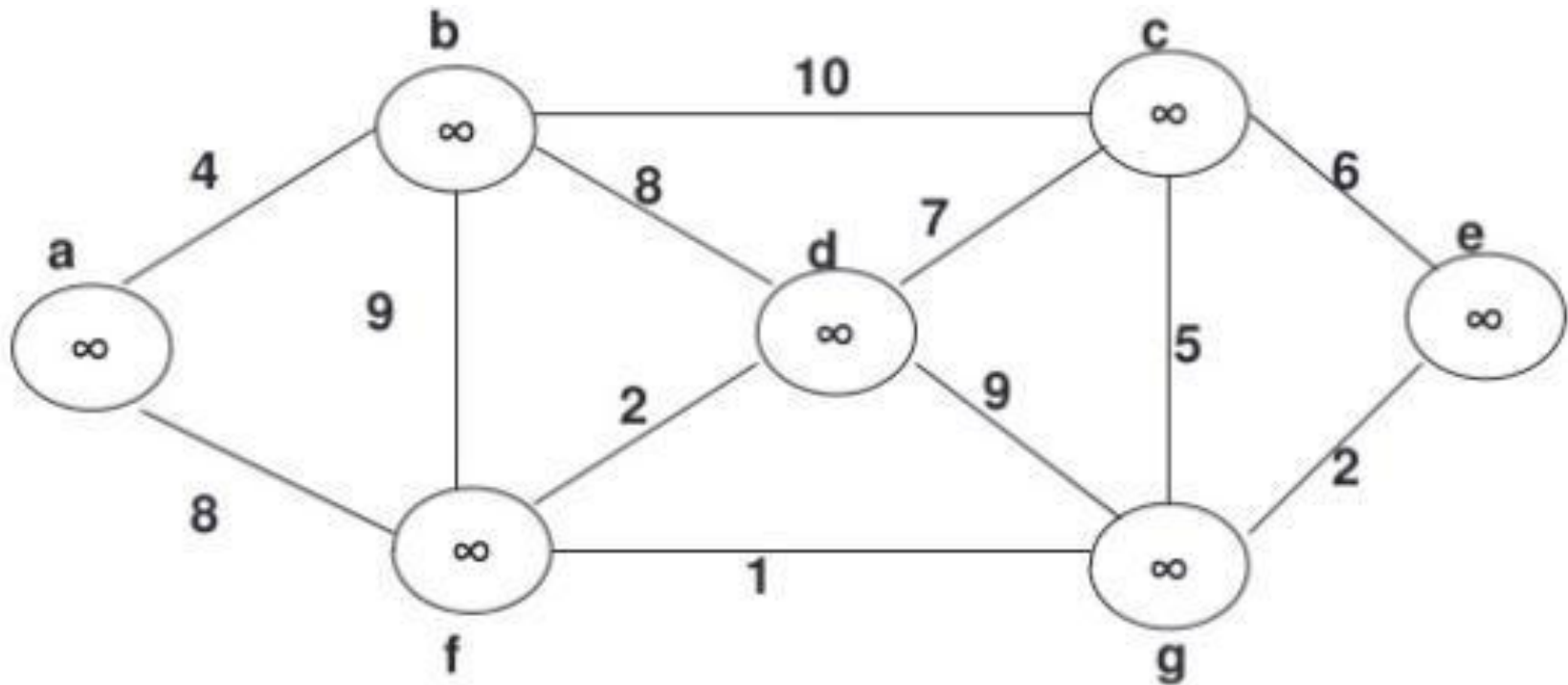
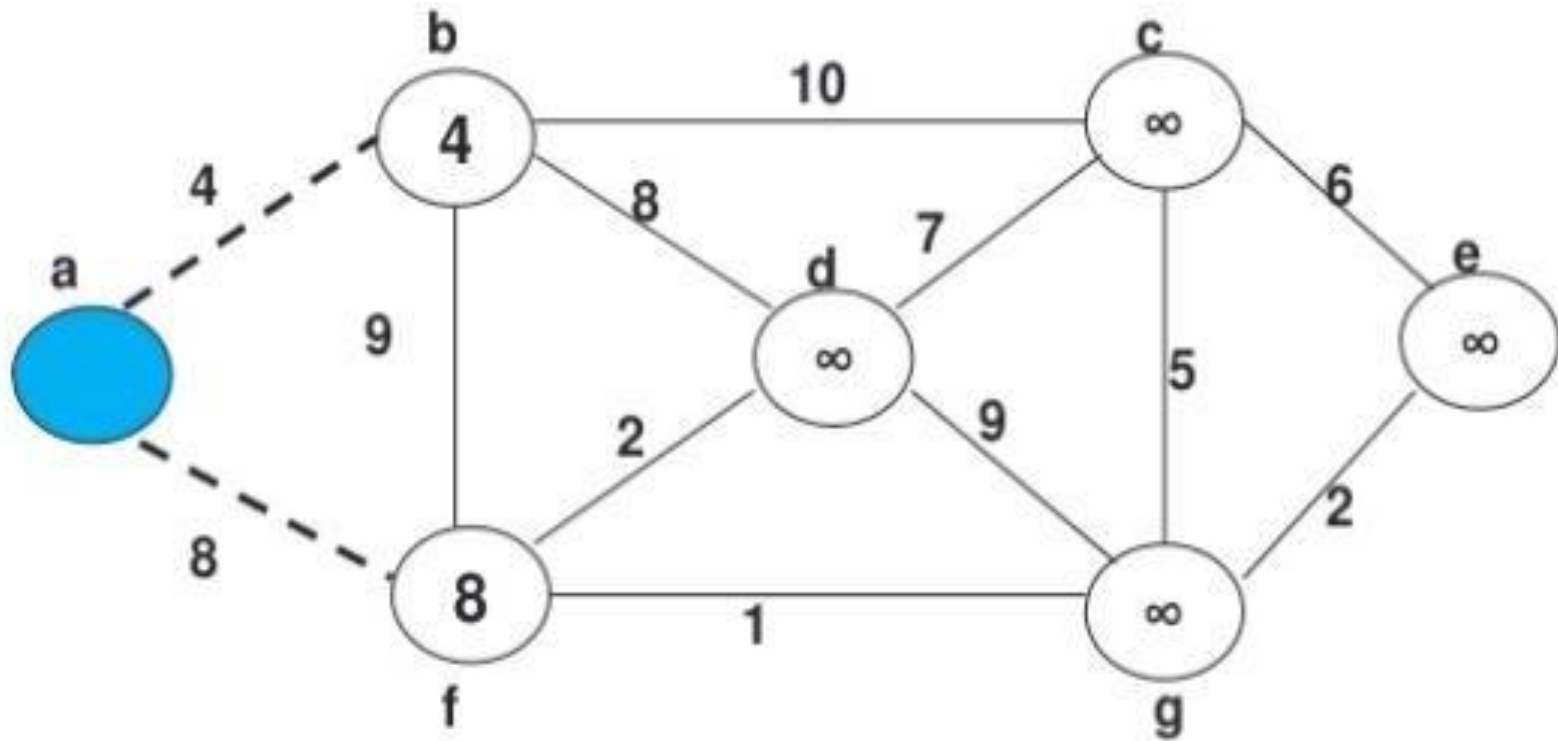
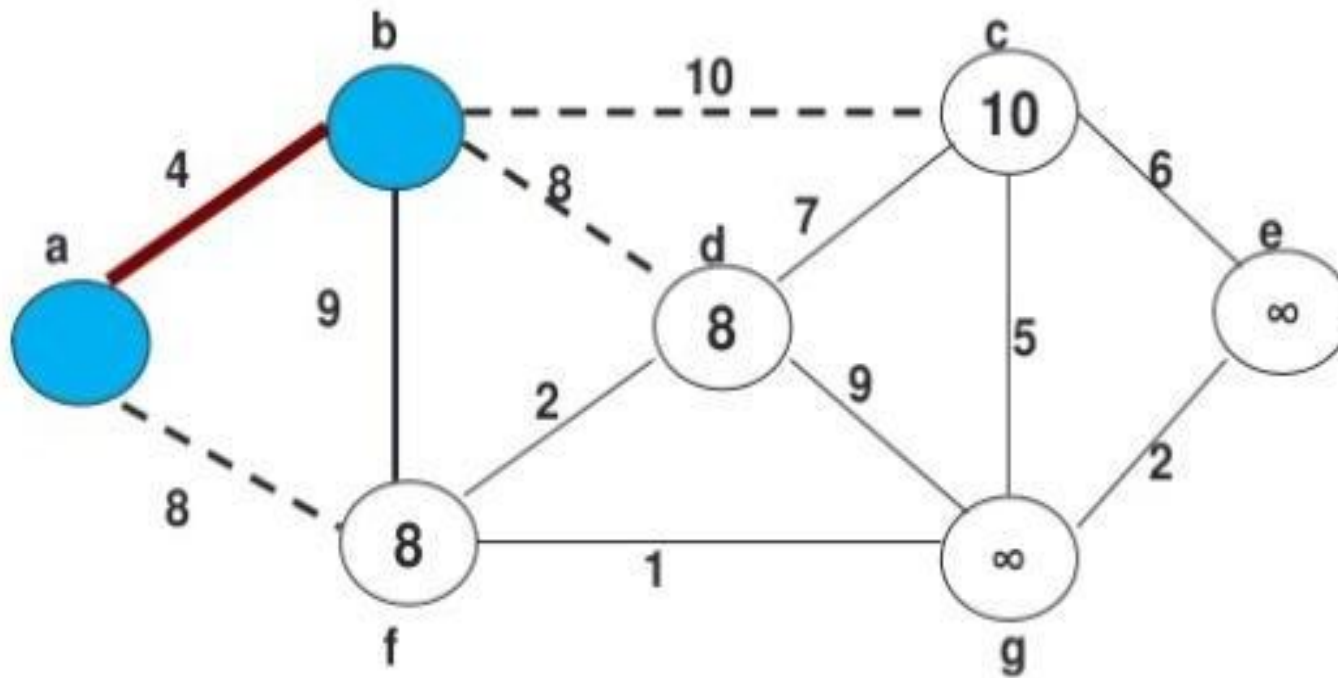
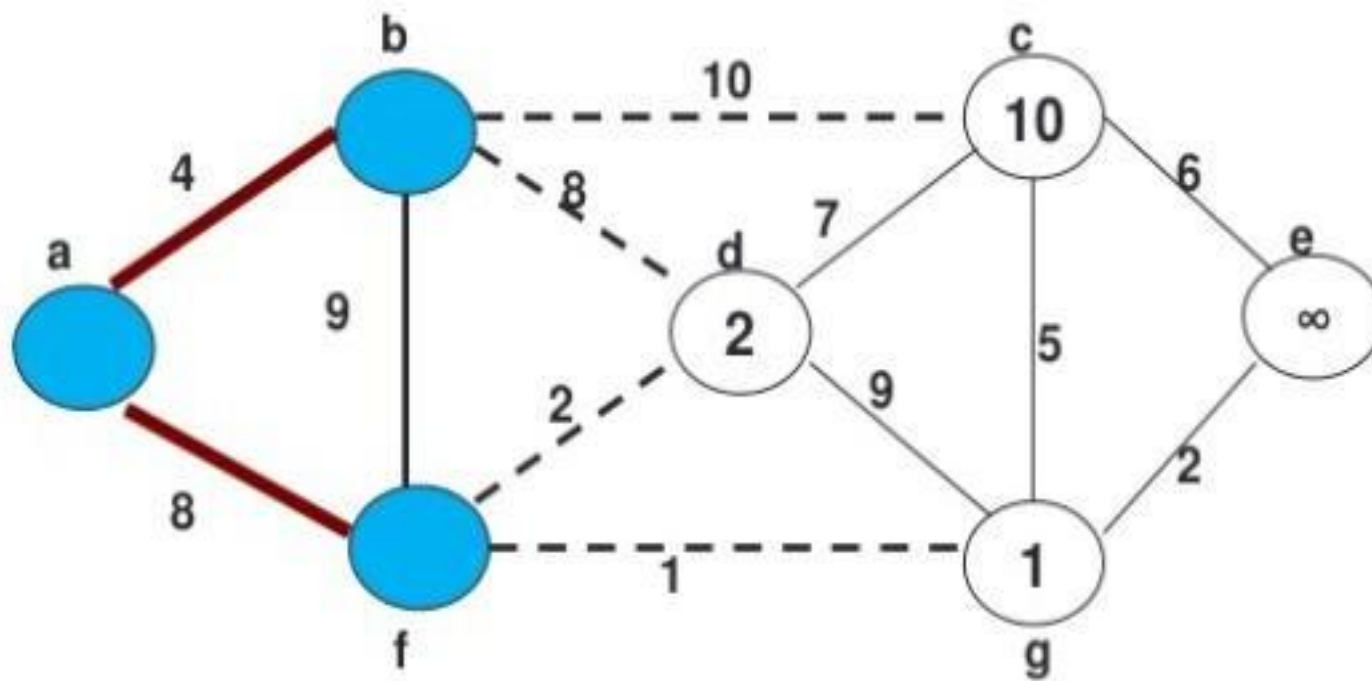


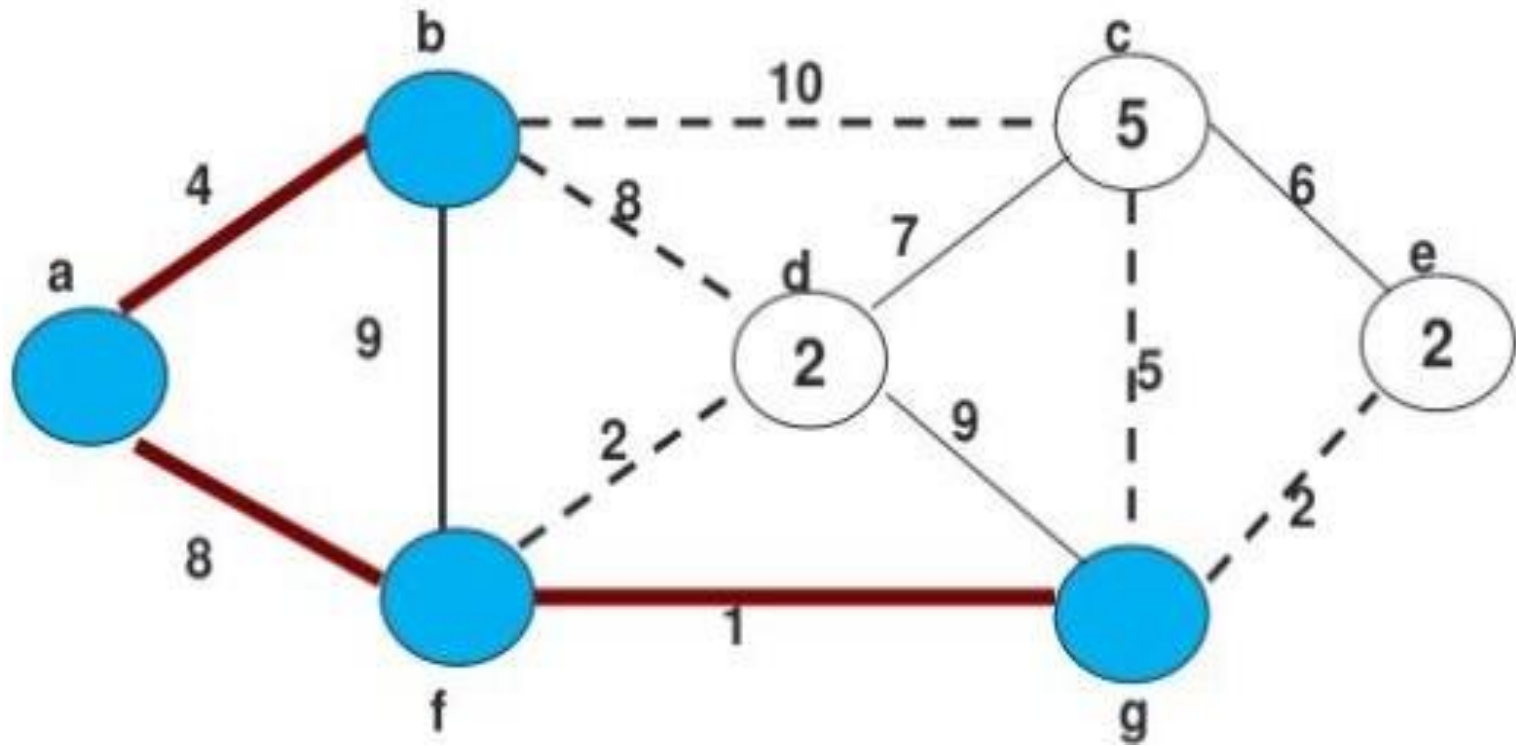
Find the minimum spanning tree using Prim's algorithm

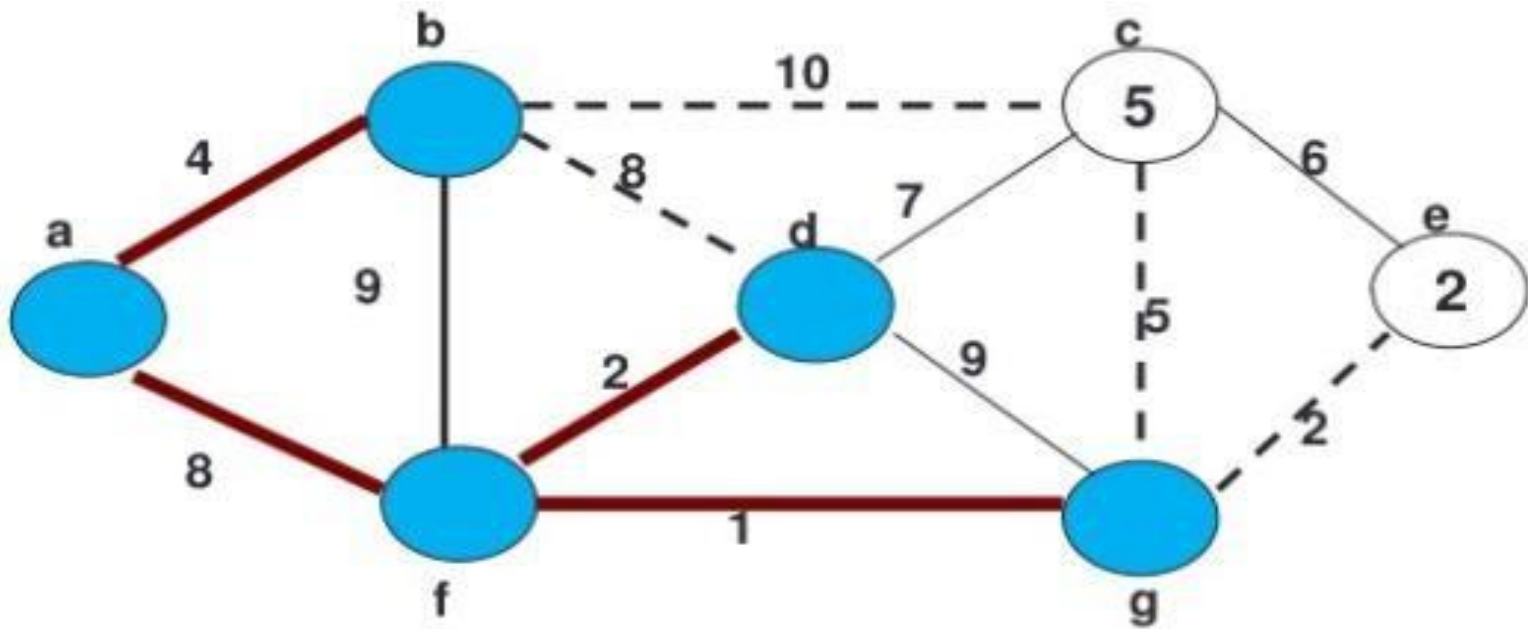


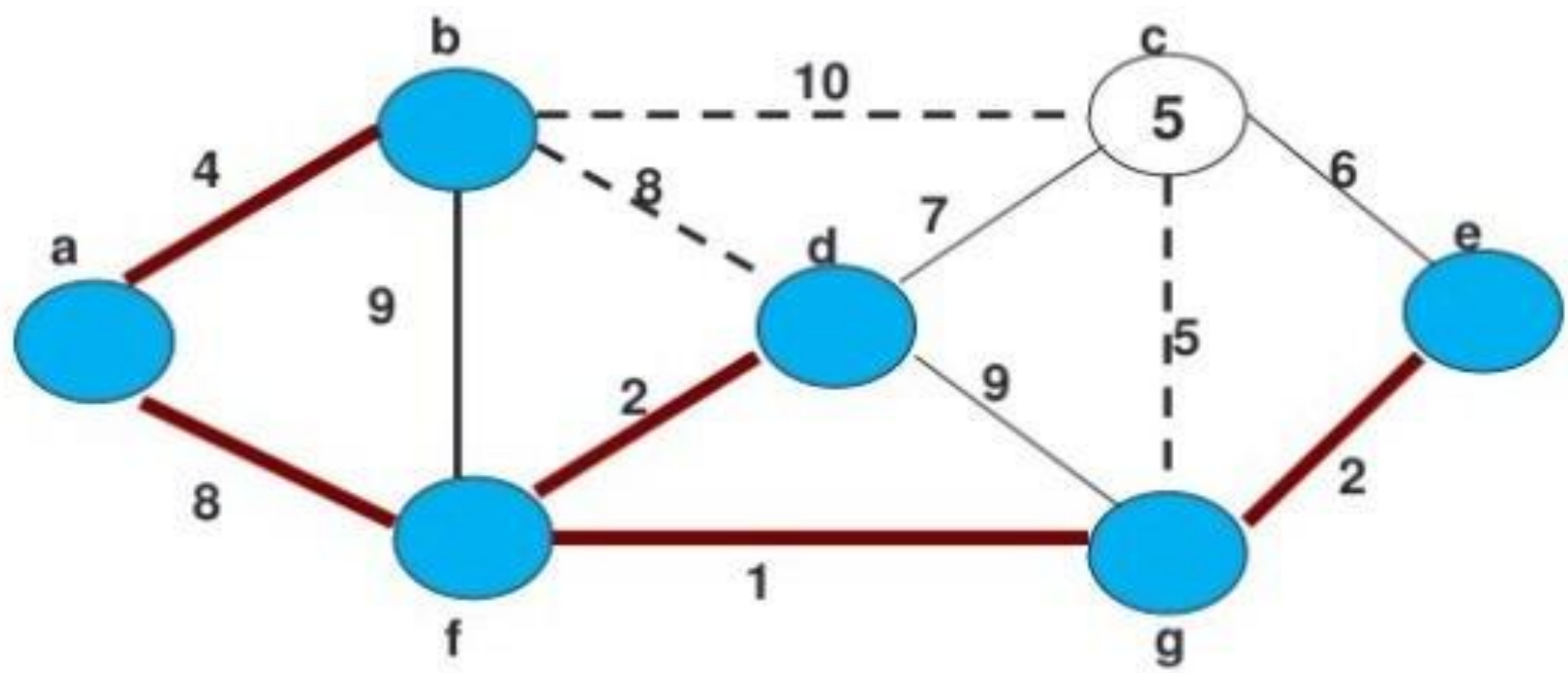




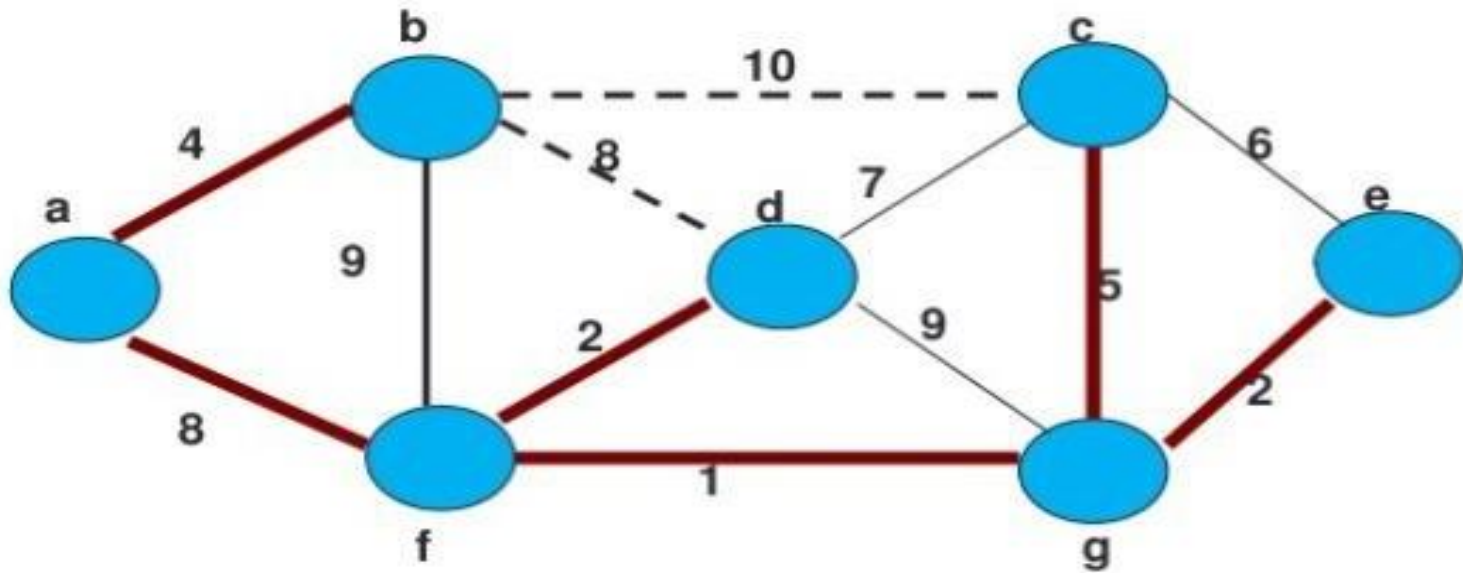






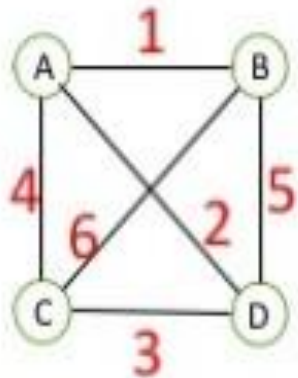


So the minimum weight is 22

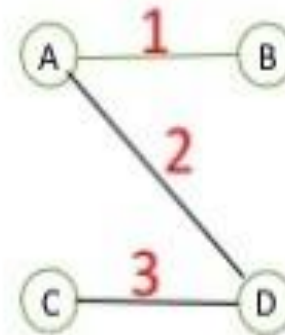


## Kruskal's Algorithm

- It is a algorithm used to find a minimum cost spanning tree for connected weighted undirected graph
- This algorithm first appeared in Proceedings of the American Mathematical Society in 1956, and was written by Joseph Kruskal

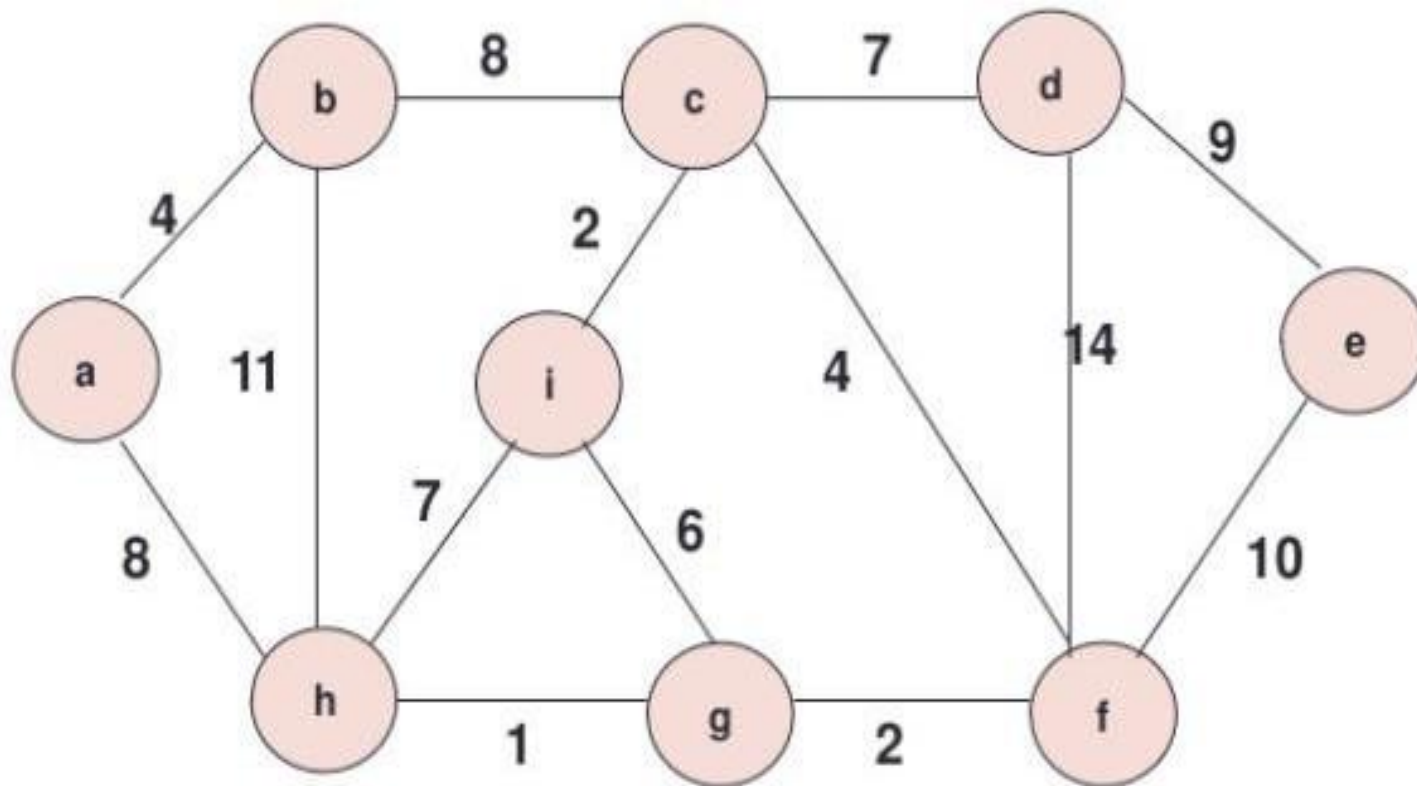


Connected Weighted



- It's a spanning tree because it connects all vertices without loops.
- Tree weight is minimum of all possibilities hence minimum cost spanning tree

Find minimum spanning tree using Kruskal's algorithm



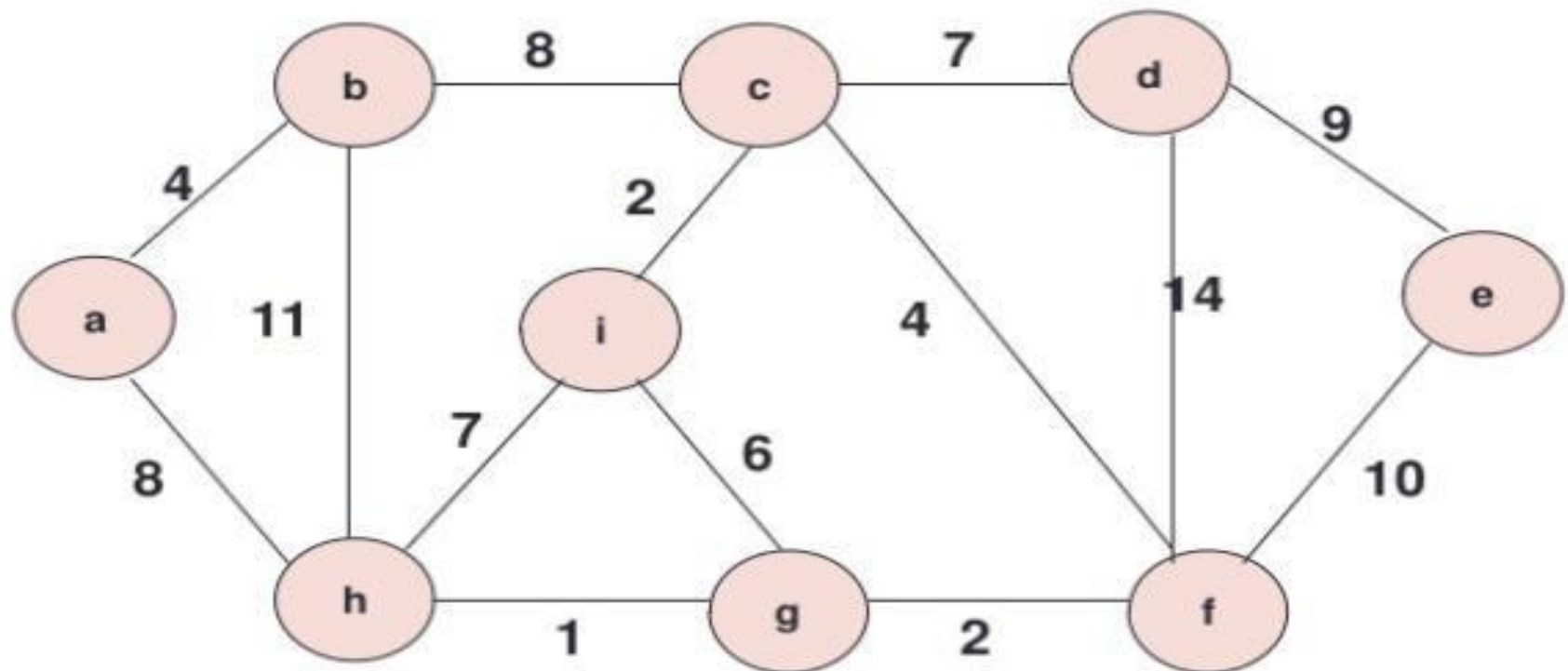
# Kruskal's Algorithm

## Step 1 : Create the edge table

- An edge table will have name of all the edges along with their weight in ascending order.
- Look at your graph and calculate the number of edges in your graph.
- And draw a table according to number of edges.
- In our case our graph contain 13 edges.
- So our edge table will have 13 columns.

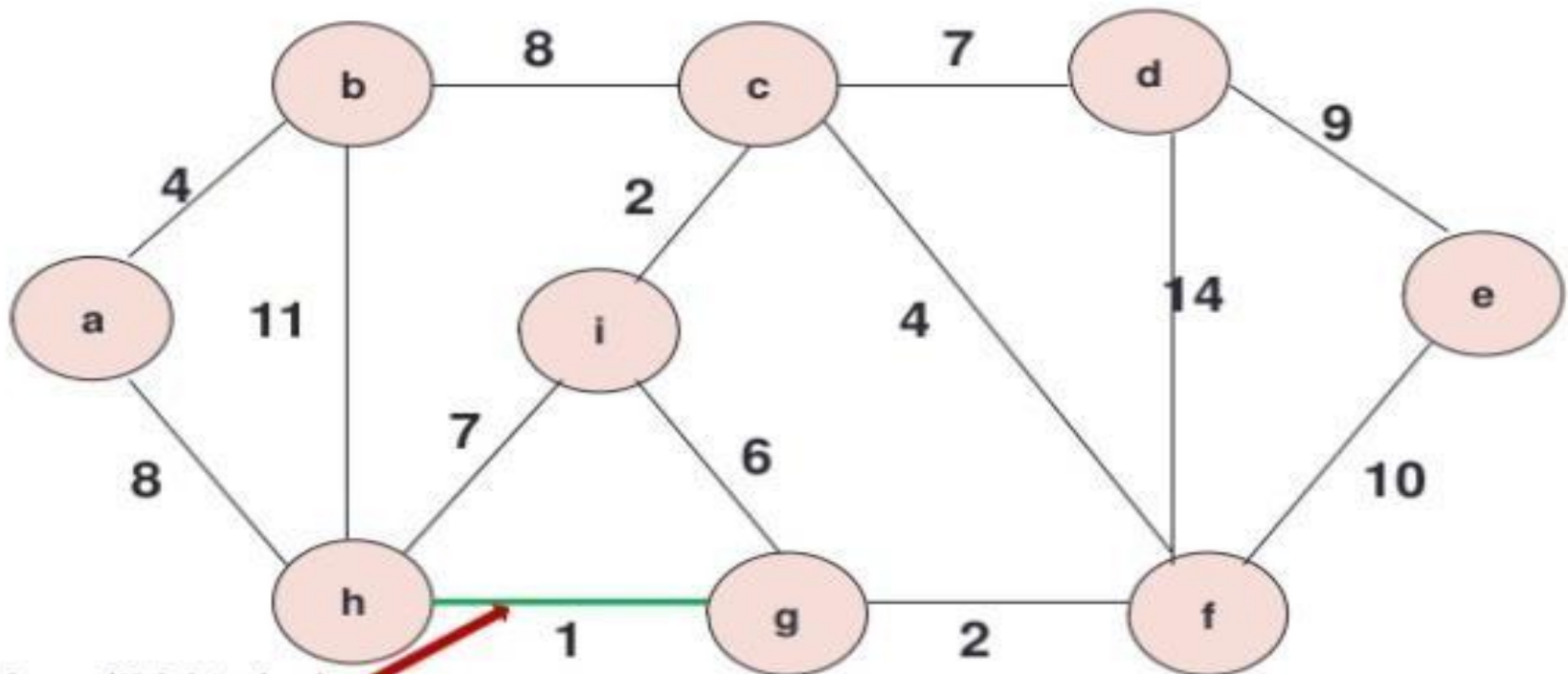
| Edges  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| Weight |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

**Fill the 1<sup>st</sup> Column with the edge of minimum weight**



|        |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--------|----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| Edges  | hg |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Weight | 1  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

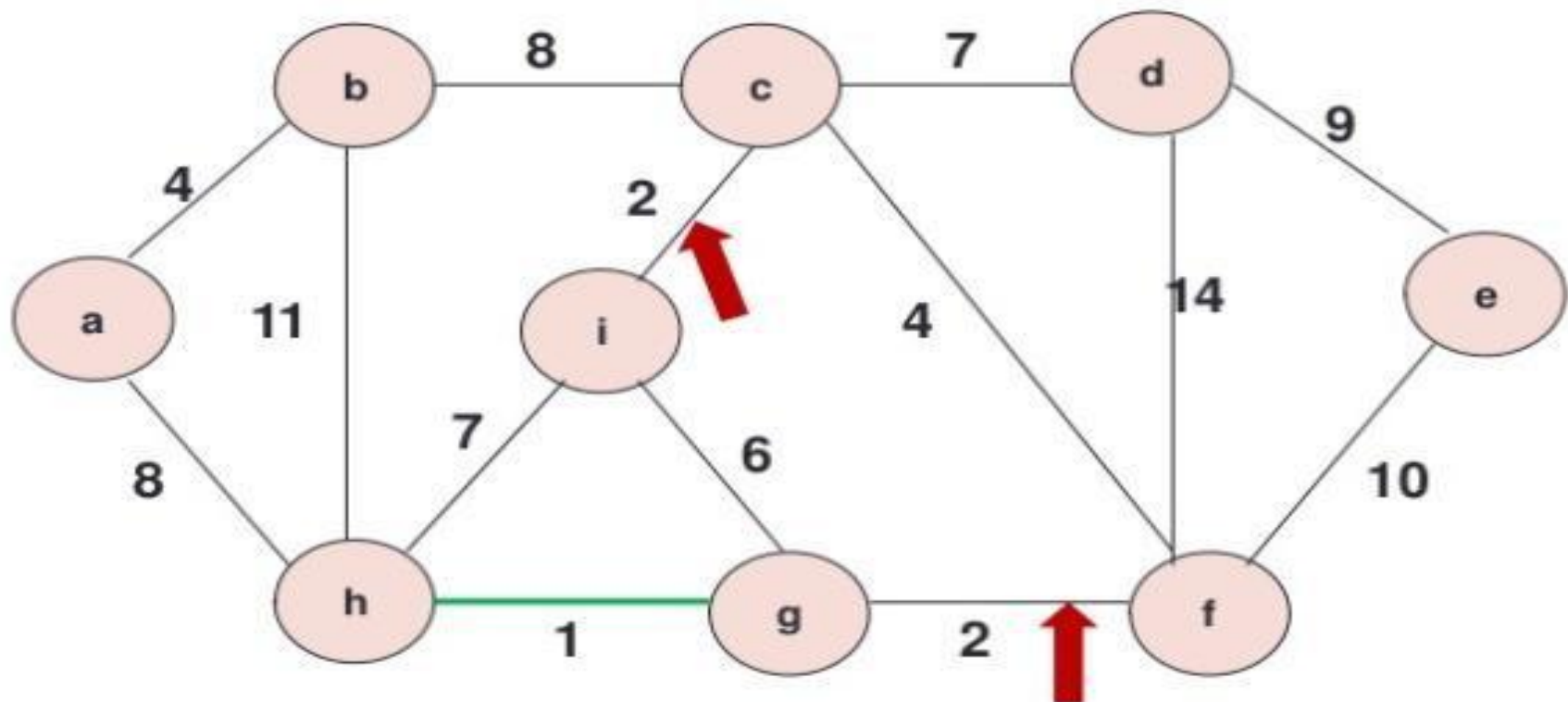
Fill the 1<sup>st</sup> Column with the edge of minimum weight



Minimum Weight edge is

|        |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--------|----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| Edges  | hg |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Weight | 1  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

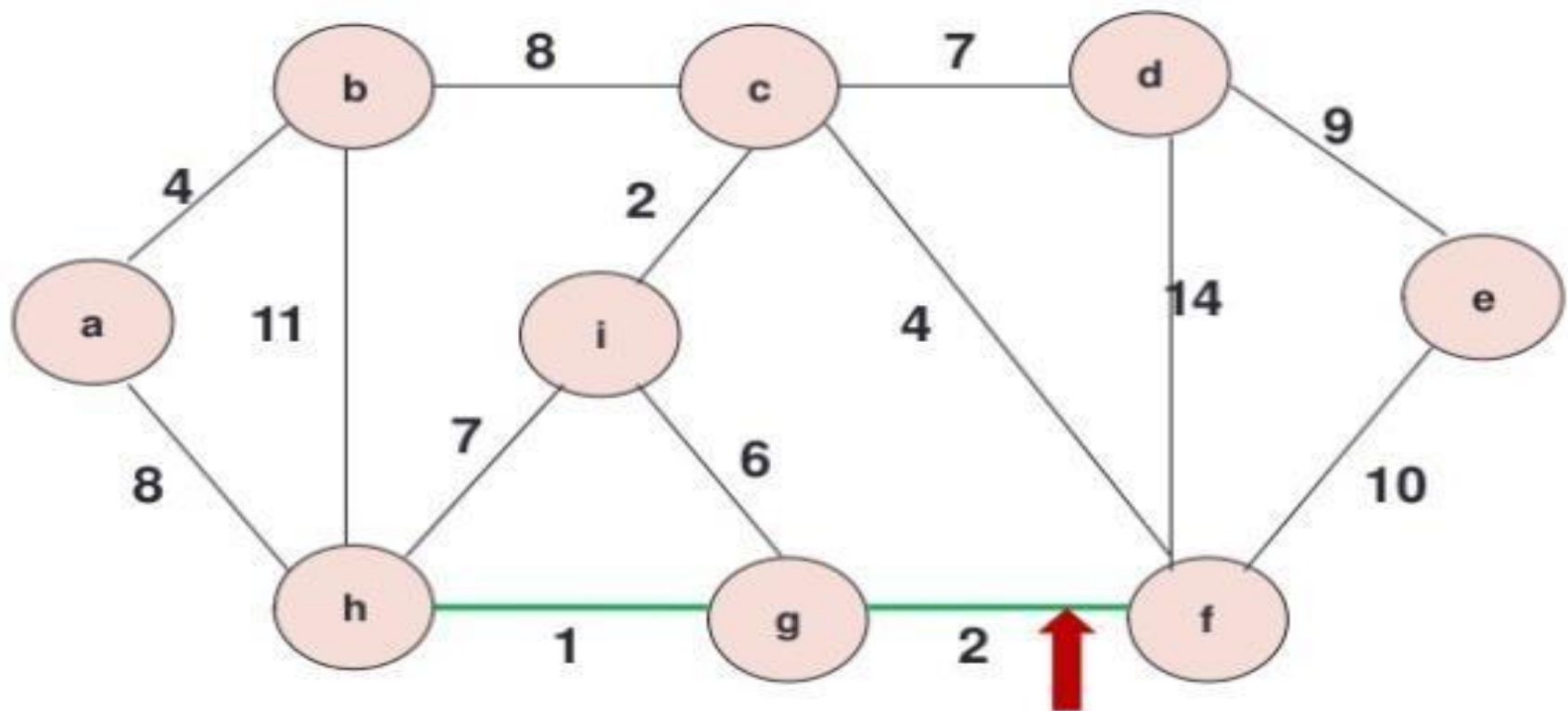
Fill the 2<sup>nd</sup> Column with the edge of minimum weight



There is two edges with same weight so select any one first

|        |    |    |  |  |  |  |  |  |  |  |  |  |  |  |
|--------|----|----|--|--|--|--|--|--|--|--|--|--|--|--|
| Edges  | hg | gf |  |  |  |  |  |  |  |  |  |  |  |  |
| Weight | 1  | 2  |  |  |  |  |  |  |  |  |  |  |  |  |

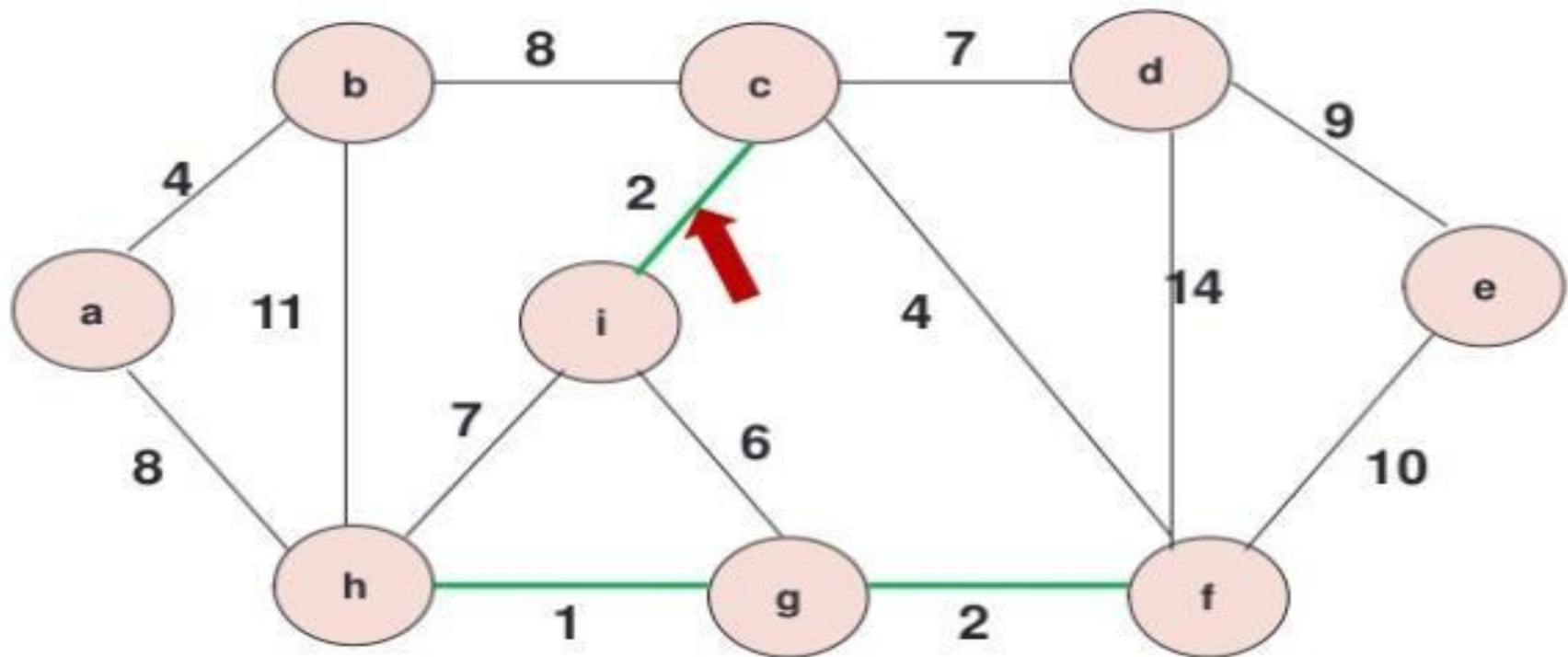
Fill the 2<sup>nd</sup> Column with the edge of minimum weight



So we Select edge "gf" first

| Edges  | hg | gf | ci |  |  |  |  |  |  |  |  |  |  |  |
|--------|----|----|----|--|--|--|--|--|--|--|--|--|--|--|
| Weight | 1  | 2  | 2  |  |  |  |  |  |  |  |  |  |  |  |

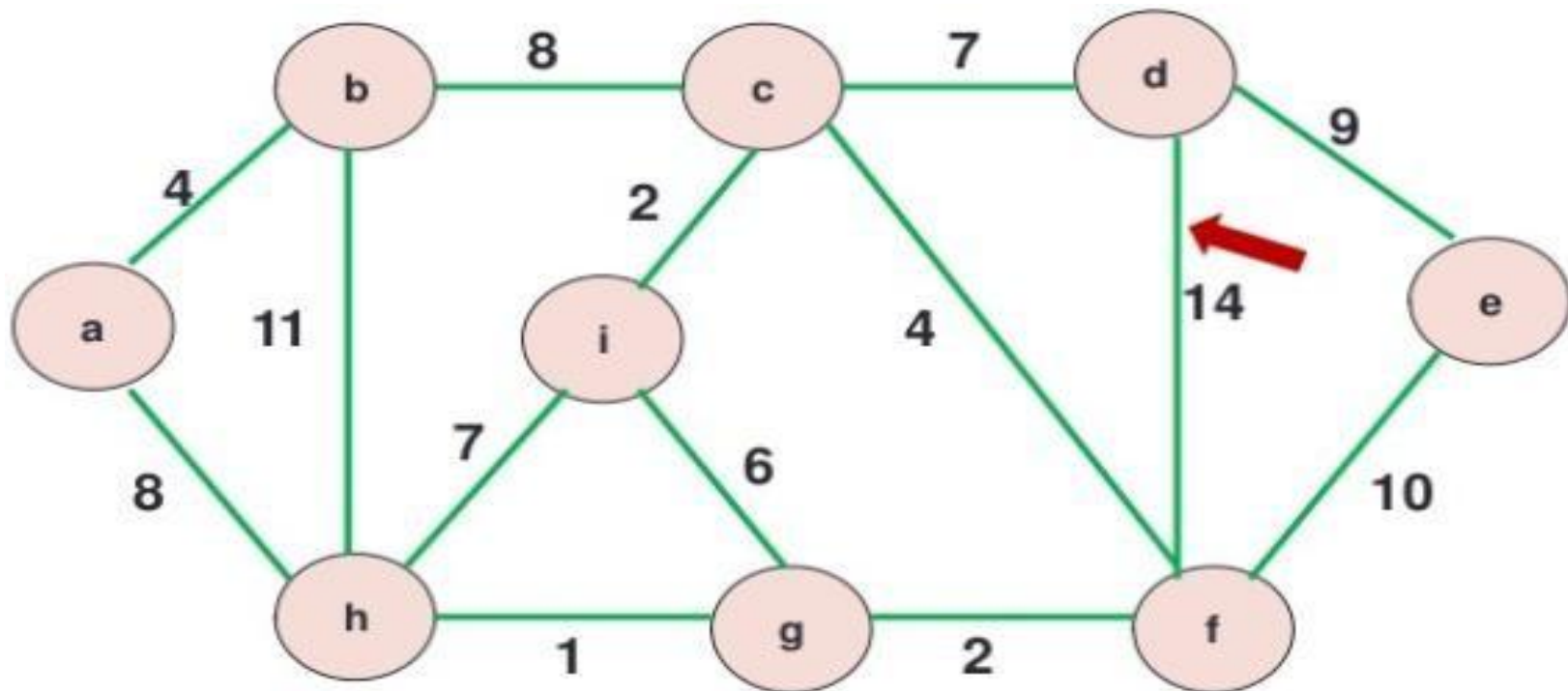
Fill the 3<sup>rd</sup> Column with the edge of minimum weight



Now Select edge "ci" and fill the column

Similarly arrange all the edges in ascending order

| Edges  | hg | gf | ci | ab | cf | ig | cd | ih | bc | ah | de | ef | bh | df |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Weight | 1  | 2  | 2  | 4  | 4  | 6  | 7  | 7  | 8  | 8  | 9  | 10 | 11 | 14 |



## Step 2

Our graph has 9 vertices so, our MST will have 8 edges.

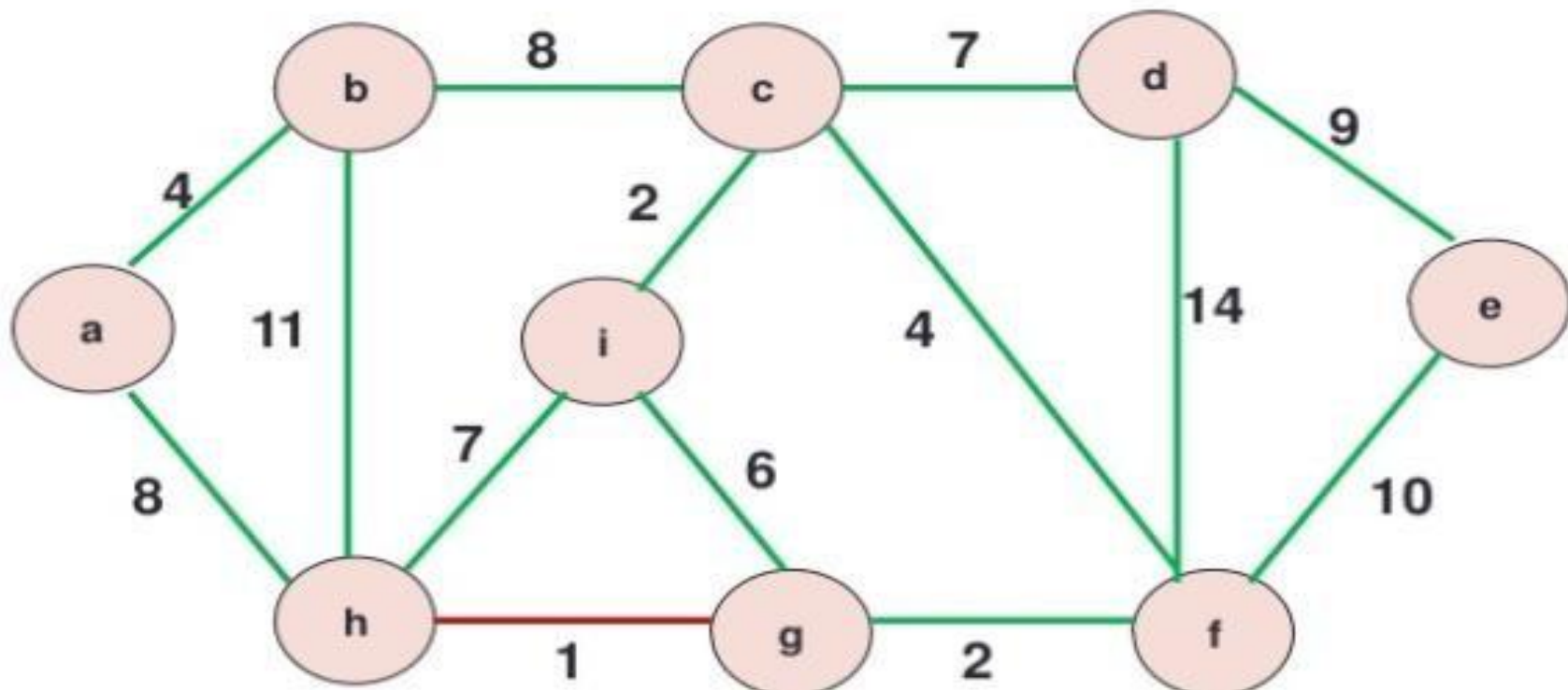
### How to find Minimum Spanning Tree

To find the Minimum Spanning Tree (MST) we will start from the smallest weight edge and keep selecting edges that does not have form any circuit (Loop/Cycle) with the previously selected edges.

| Edges  | hg | gf | ci | ab | cf | ig | cd | ih | bc | ah | de | ef | bh | df |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Weight | 1  | 2  | 2  | 4  | 4  | 6  | 7  | 7  | 8  | 8  | 9  | 10 | 11 | 14 |



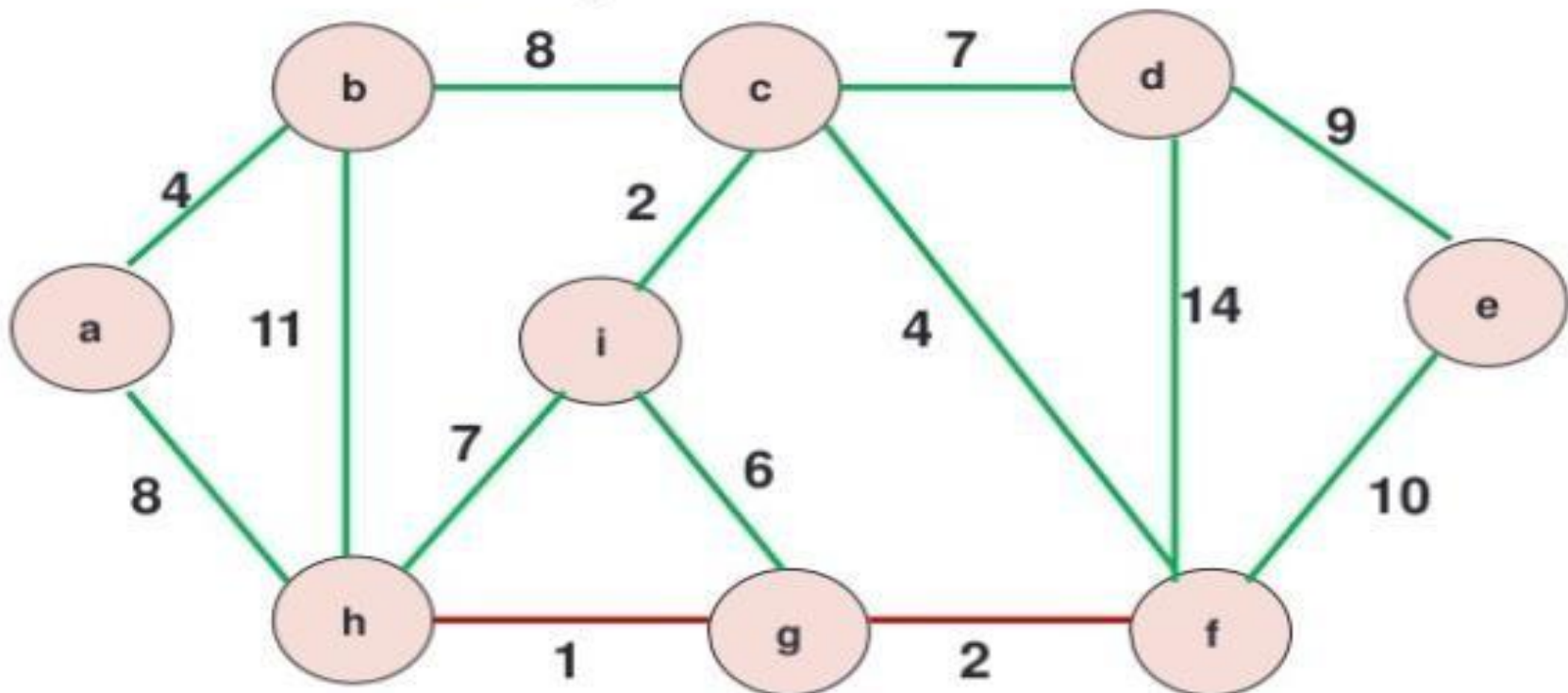
1 is the smallest weighted edge so we can select edge "hg"



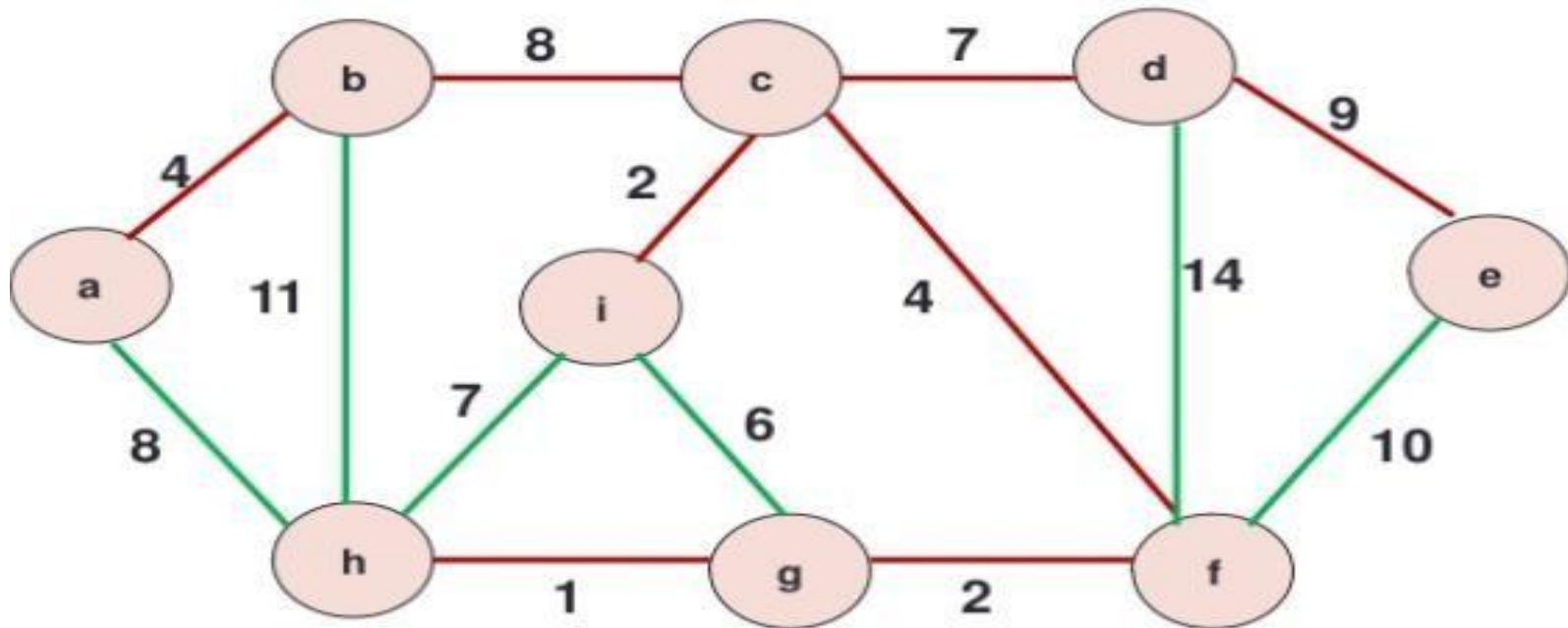
| Edges  | hg | gf | ci | ab | cf | ig | cd | ih | bc | ah | de | ef | bh | df |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Weight | 1  | 2  | 2  | 4  | 4  | 6  | 7  | 7  | 8  | 8  | 9  | 10 | 11 | 14 |



2 is the Next smallest weighted edge so we can select edge "gf" because "gf" does not make any circuit with previously selected edges

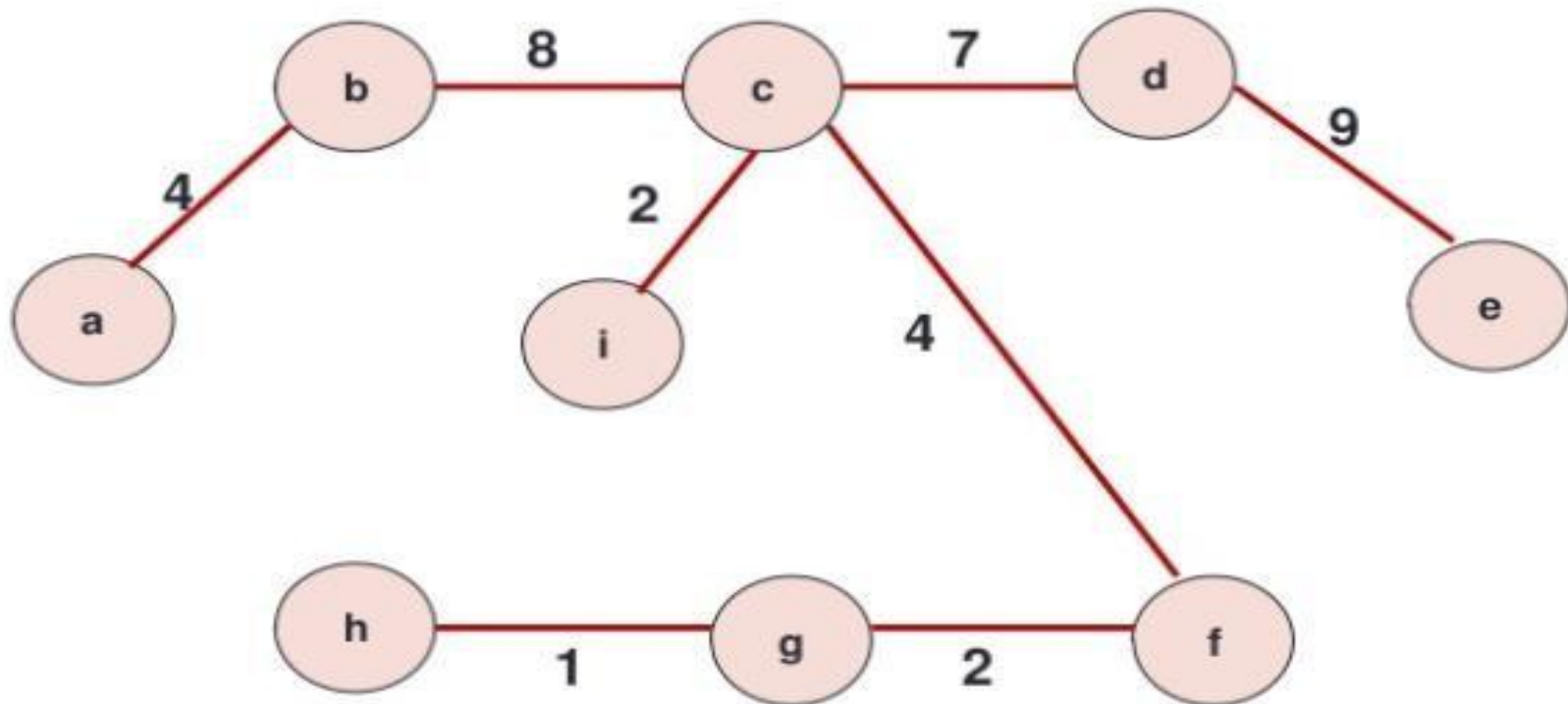


| Edges  | hg | gf | ci | ab | cf | ig | cd | ih | bc | ah | de | ef | bh | df |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Weight | 1  | 2  | 2  | 4  | 4  | 6  | 7  | 7  | 8  | 8  | 9  | 10 | 11 | 14 |

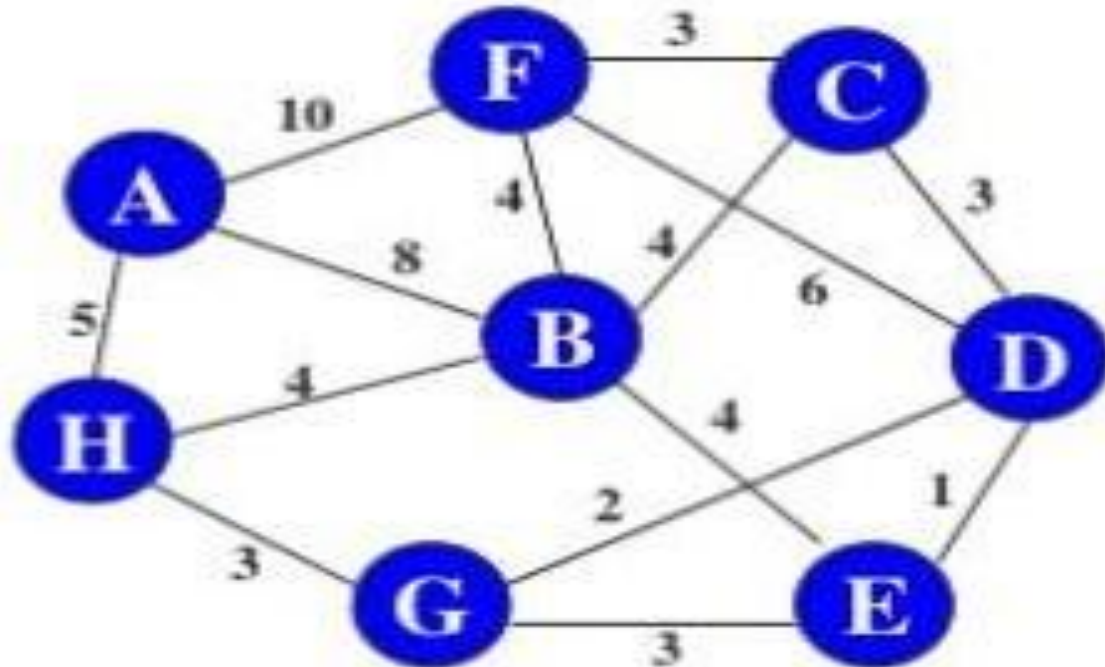


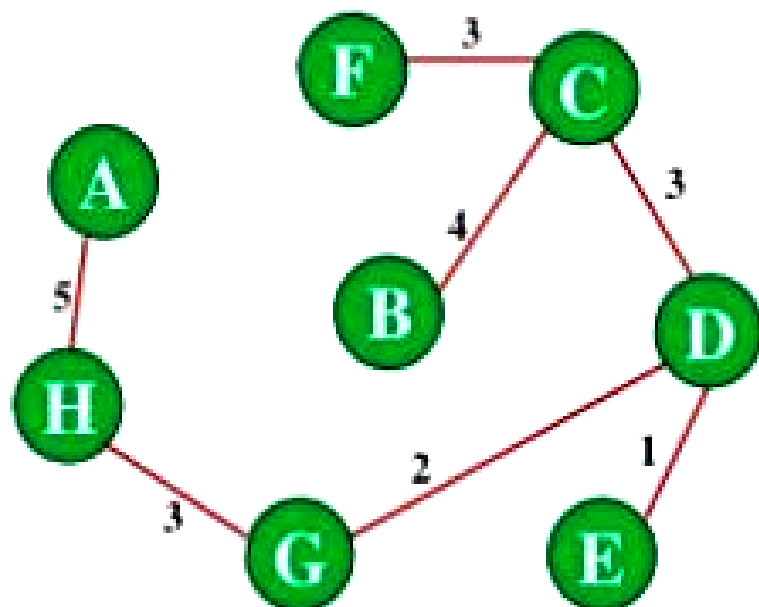
**So MST IS**

**Cost is 37**



Find minimum spanning tree using Kruskal's algorithm





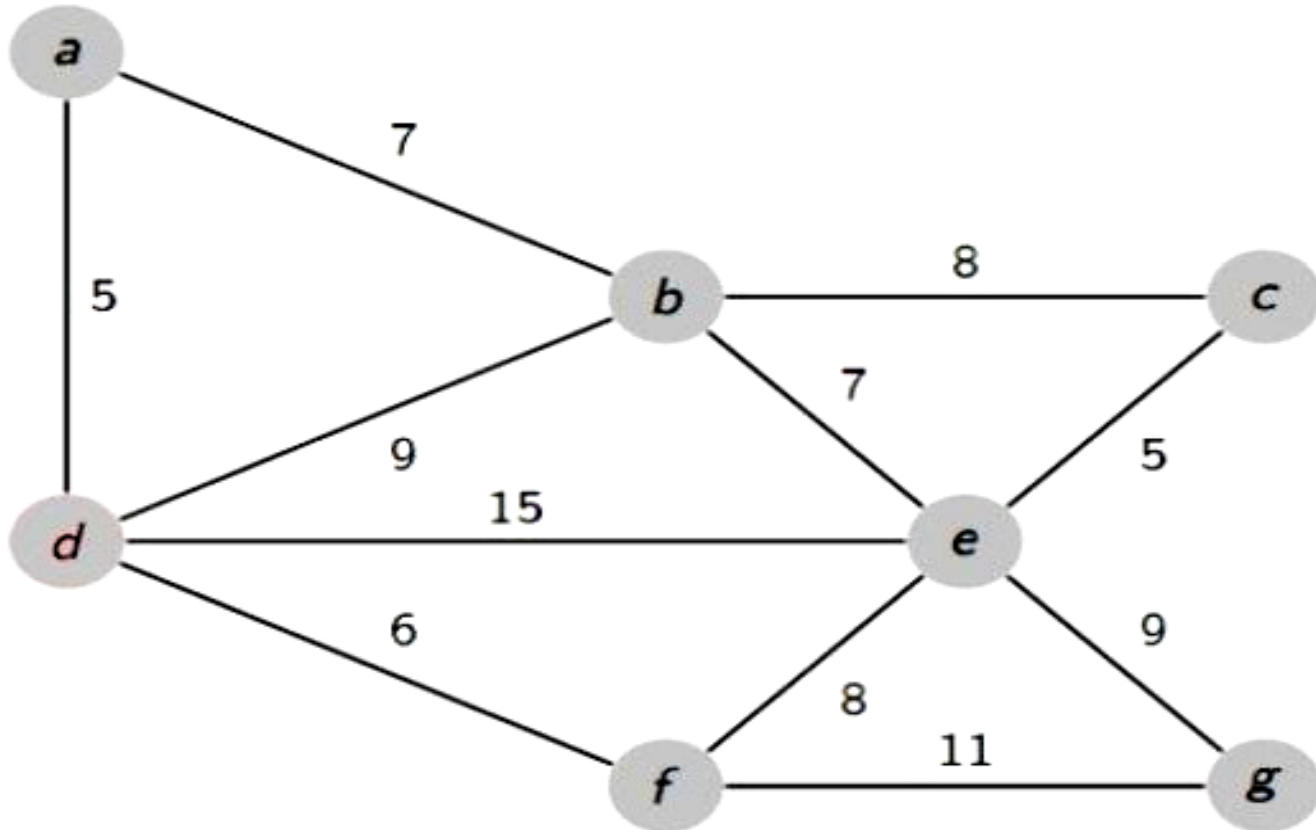
| edge  | $d_v$ |   |
|-------|-------|---|
| (D,E) | 1     | ✓ |
| (D,G) | 2     | ✓ |
| (E,G) | 3     | ✗ |
| (C,D) | 3     | ✓ |
| (G,H) | 3     | ✓ |
| (C,F) | 3     | ✓ |
| (B,C) | 4     | ✓ |

| edge  | $d_v$ |   |
|-------|-------|---|
| (B,E) | 4     | ✗ |
| (B,F) | 4     | ✗ |
| (B,H) | 4     | ✗ |
| (A,H) | 5     | ✓ |
| (D,F) | 6     | ✗ |
| (A,B) | 8     | ✗ |
| (A,F) | 10    | ✗ |

**Done!**

$$\text{Total Cost} = \sum d_v = 21$$

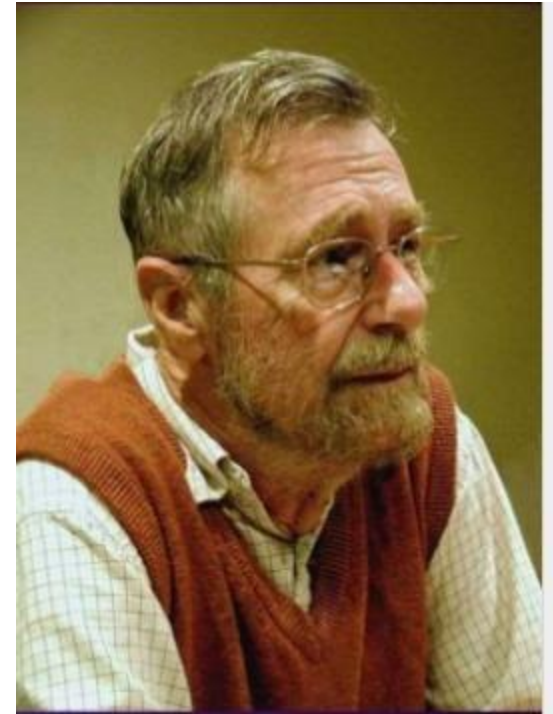
Find the minimum spanning tree using Prim's algorithm.



# Shortest Path Problem

## Dijkstra's Algorithm

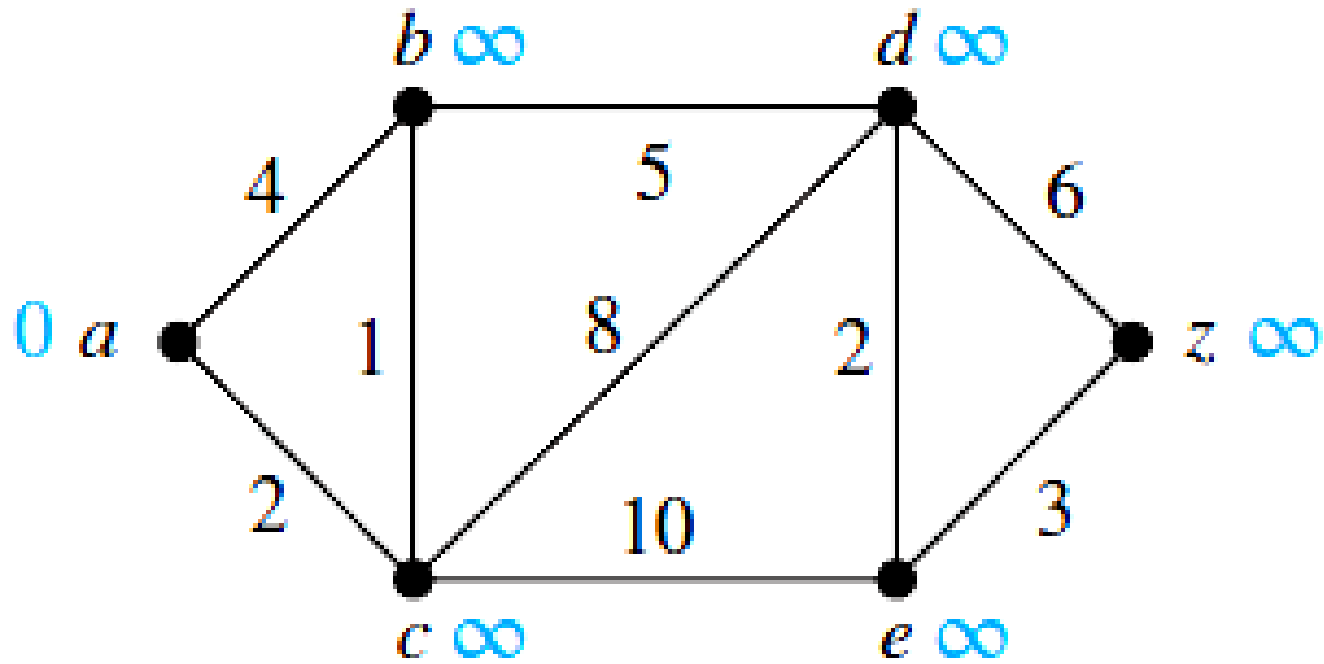
Dijkstra's Algorithm  
derived by Dutch  
computer scientist  
Edgser Wybe Dijkstra  
in 1956 and published  
in 1956.



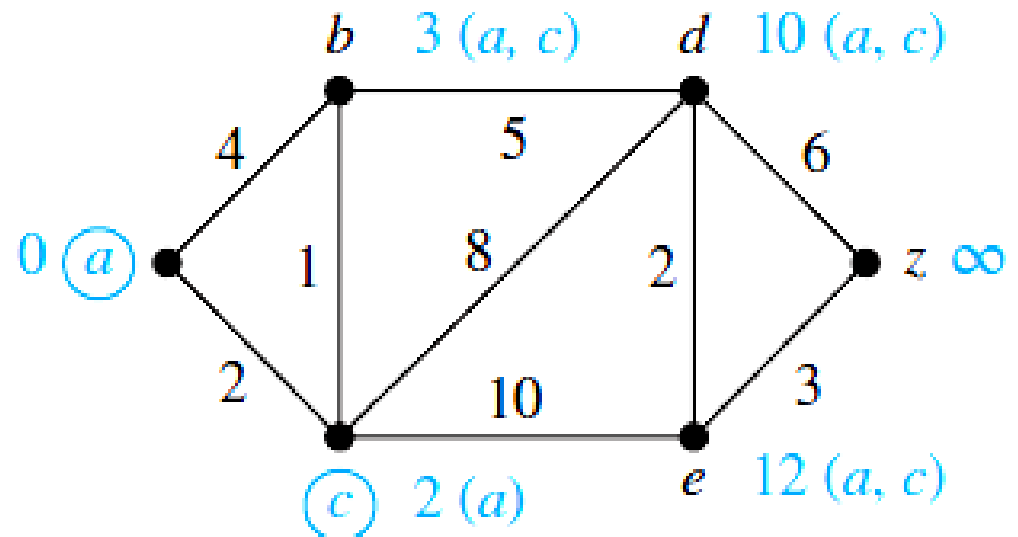
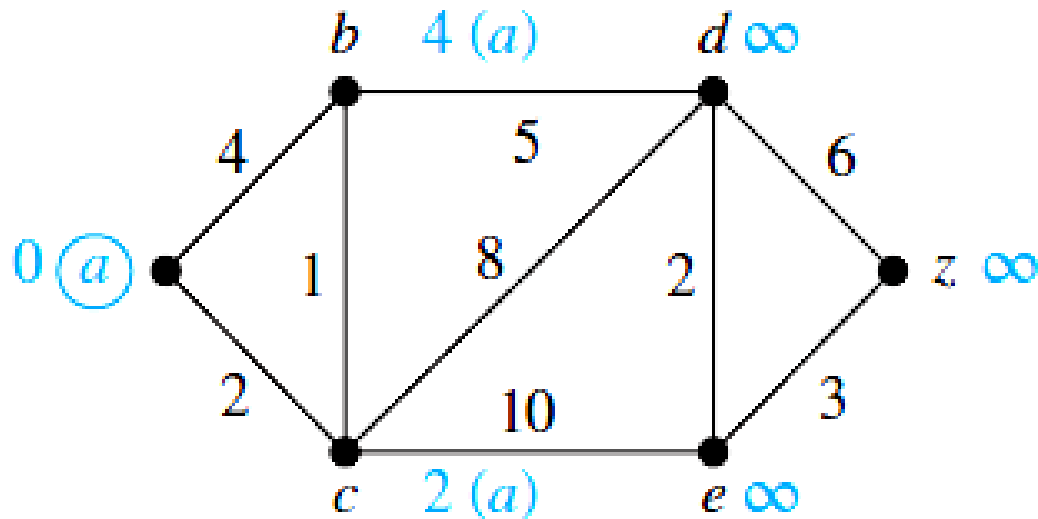
# Introduction



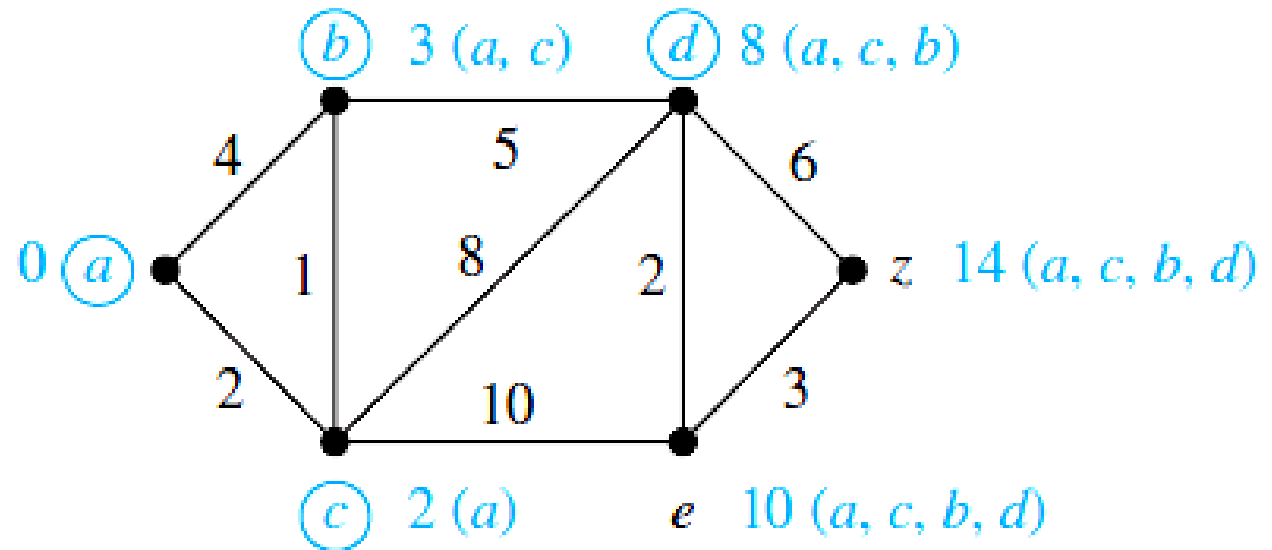
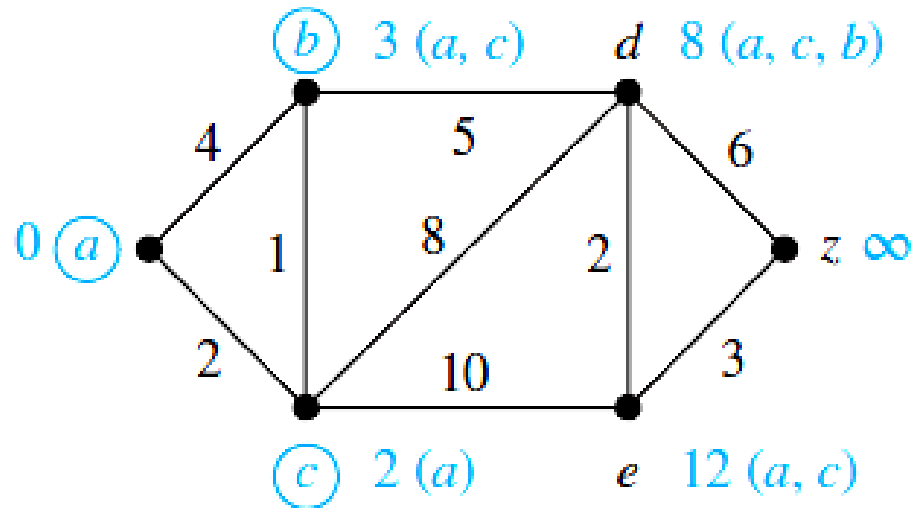
- Dijkstra's Algorithm is a solution to the single-source shortest path problem in graph theory.
- It works on both directed and undirected graphs. However, all edges must have nonnegative weights.
- Input : Weighted graph  $G=\{E,V\}$  and source vertex, such that all edge weights are nonnegative.
- Output: Length of shortest paths(or the shortest paths themselves) from a given source vertex to all other vertices.



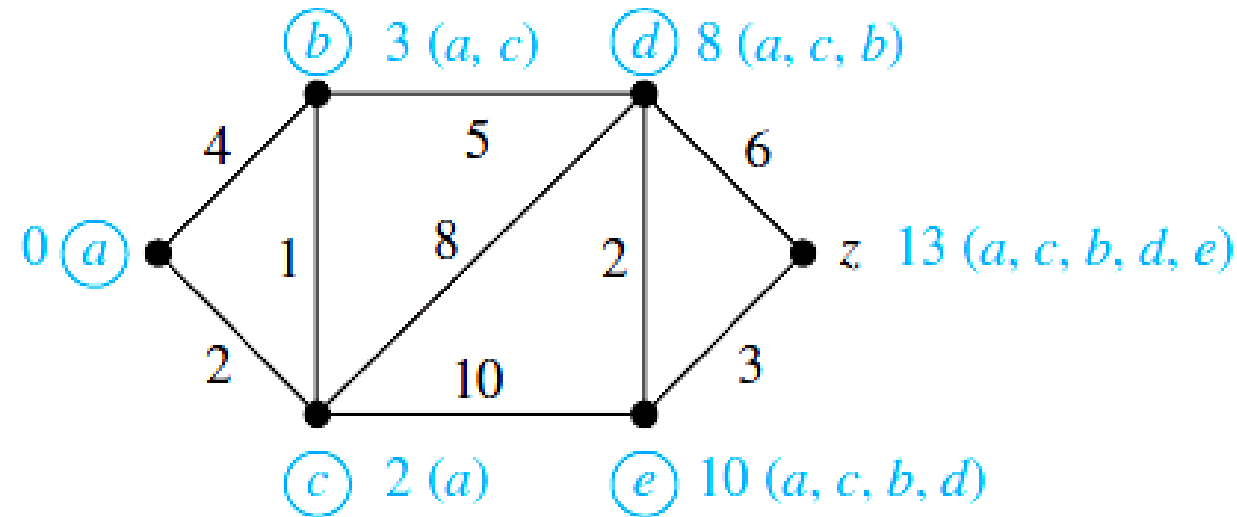
# Dijkstra's Algorithm



# Dijkstra's Algorithm



# Dijkstra's Algorithm



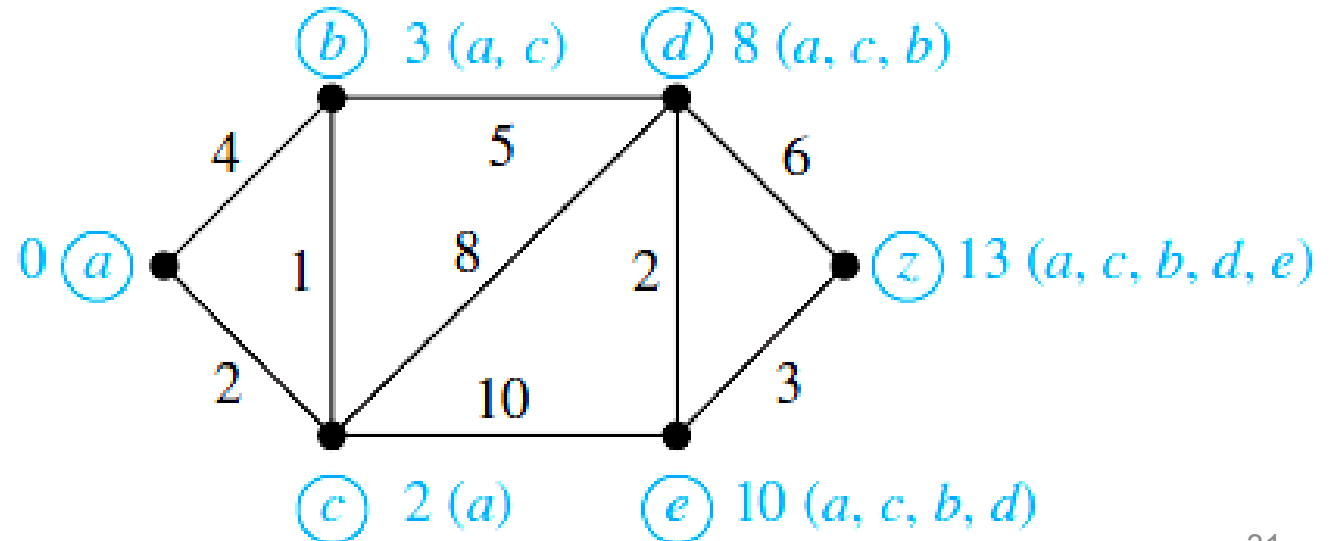
$$a - c : 2$$

$$a - c - b : 3$$

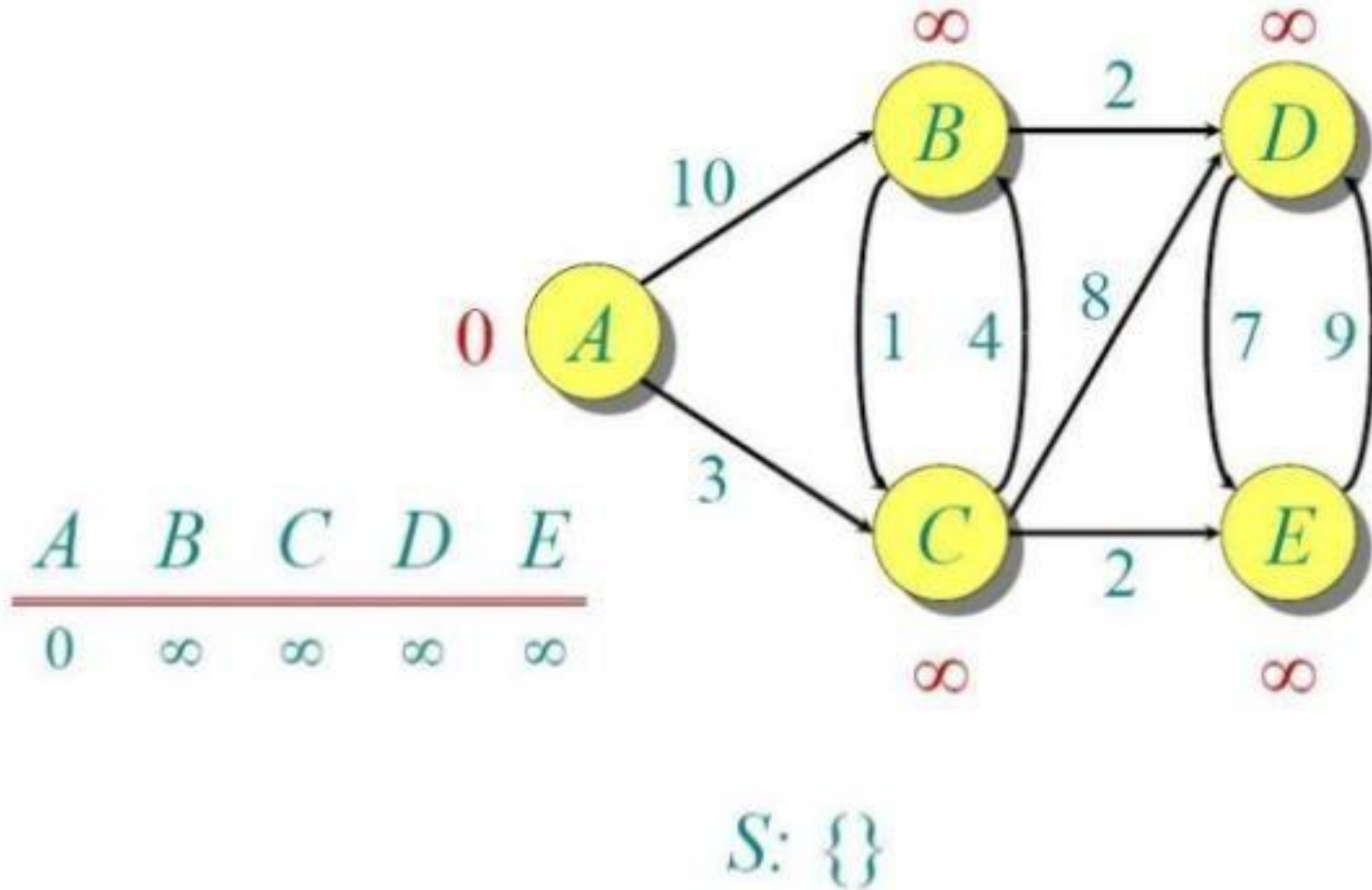
$$a - c - b - d : 8$$

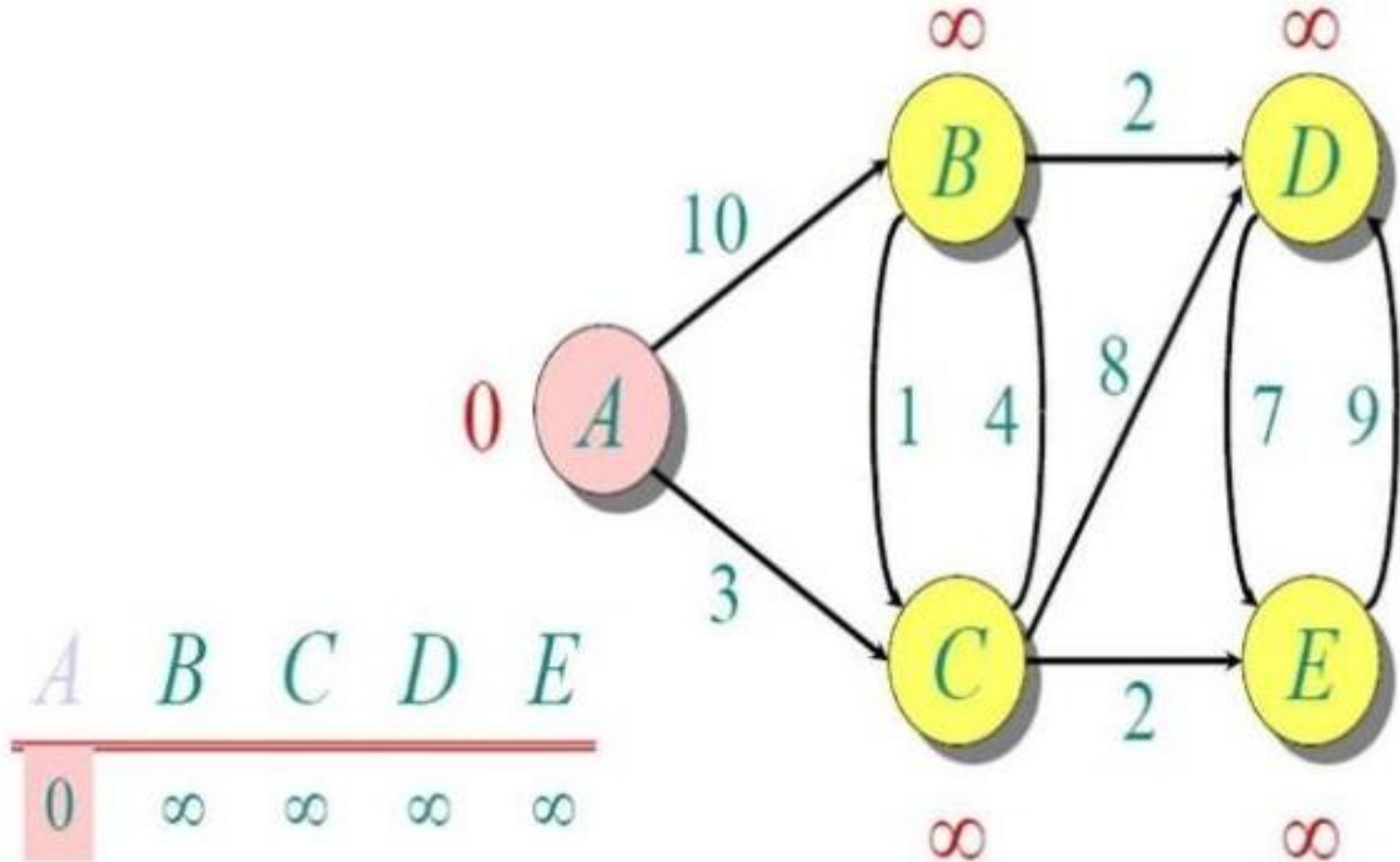
$$a - c - b - d - e : 10$$

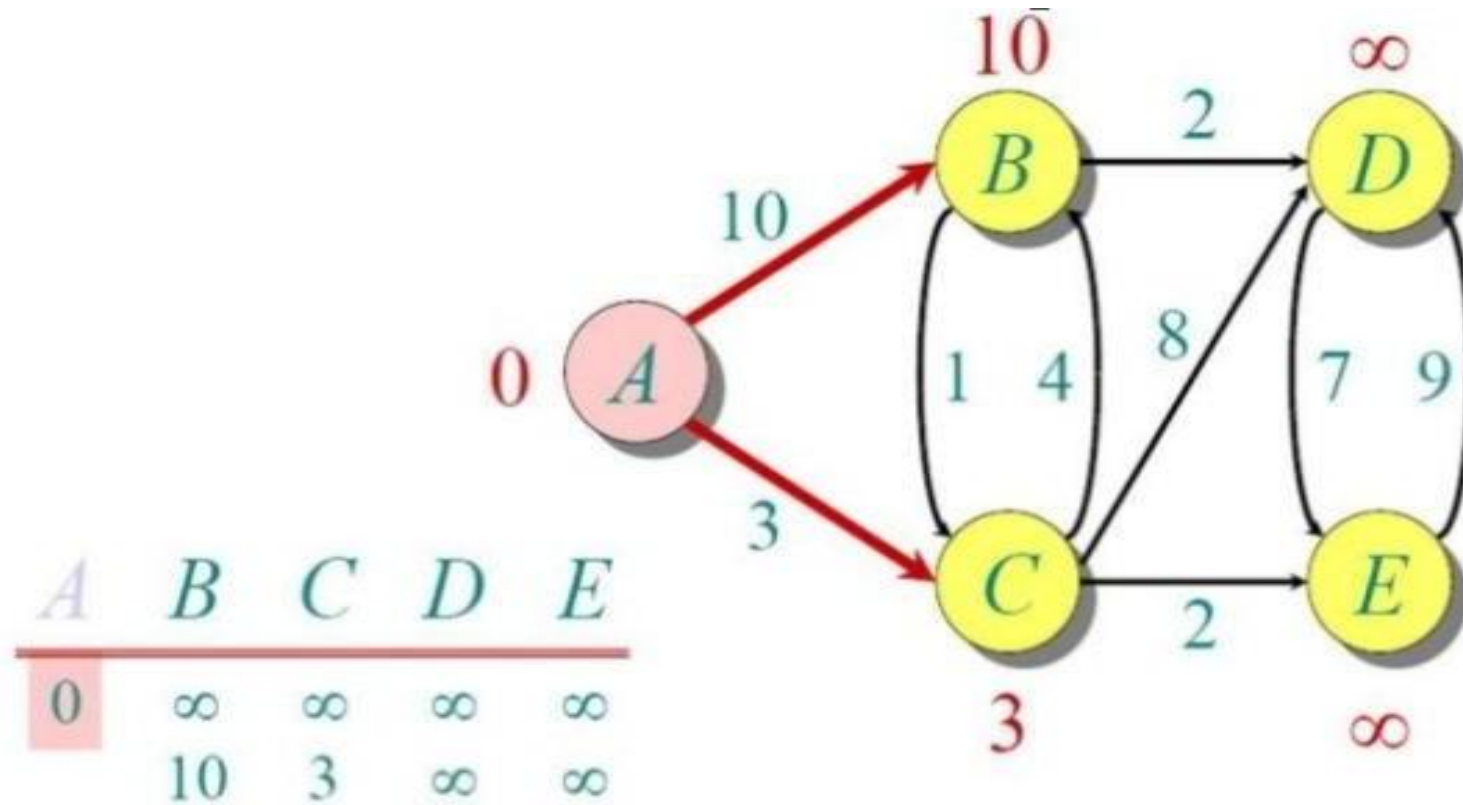
$$a - c - b - d - e - z : 13$$



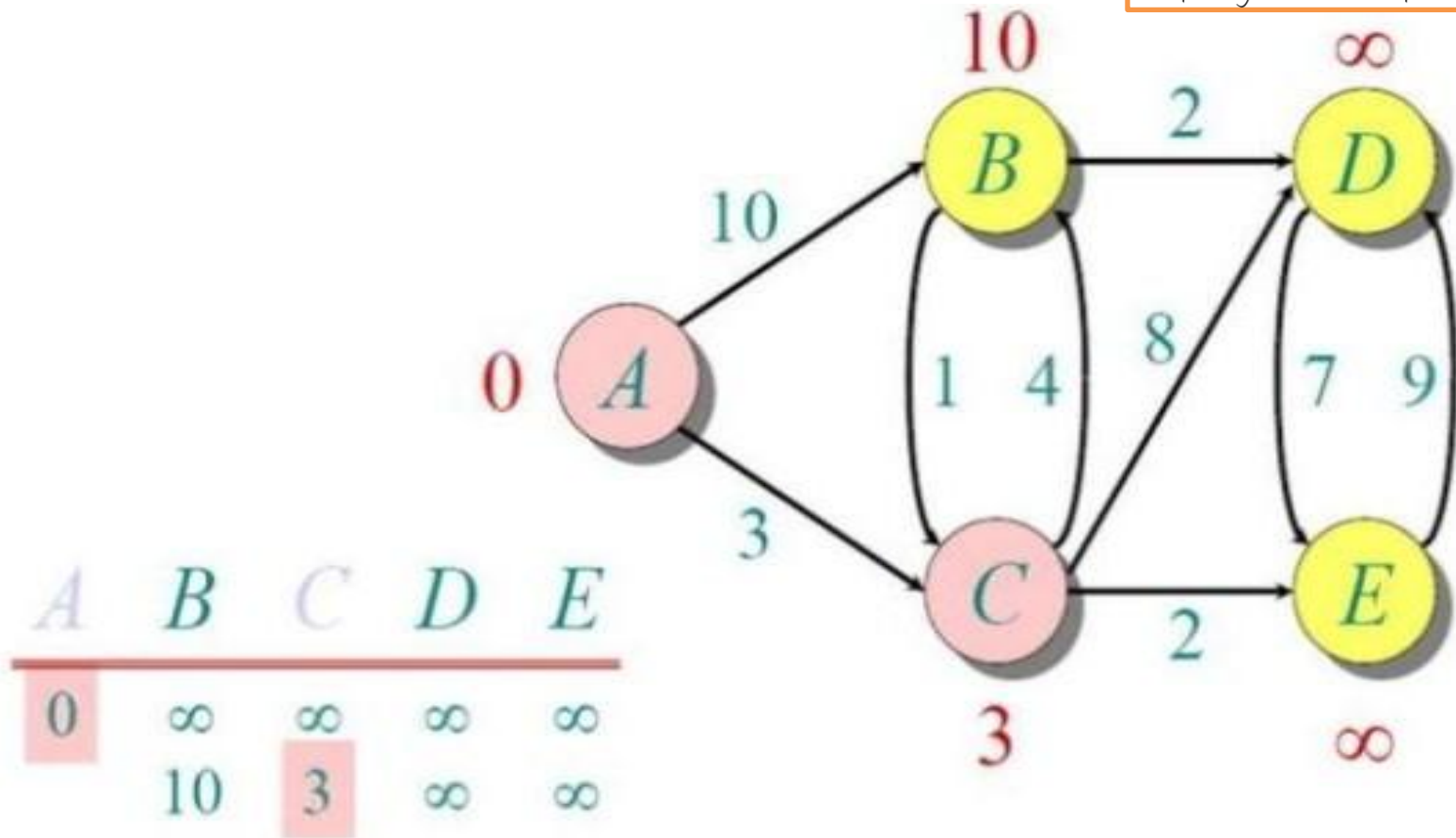
# Example



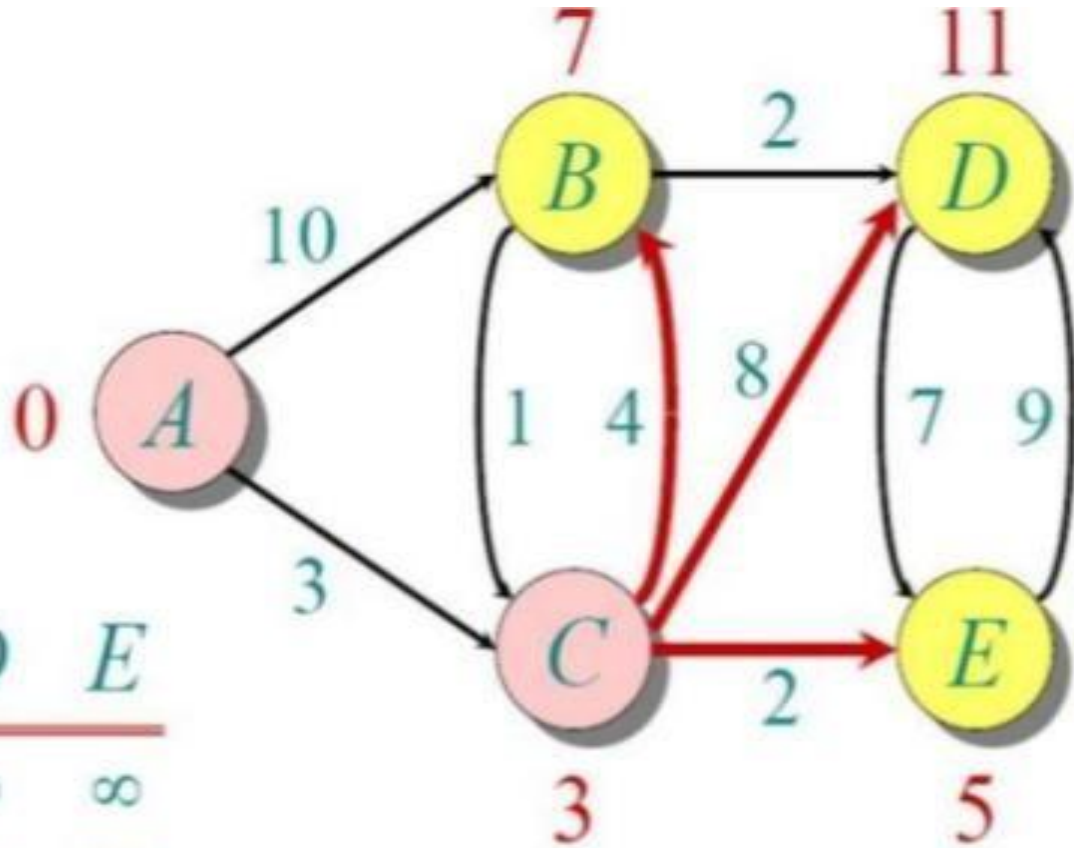




$S: \{A\}$

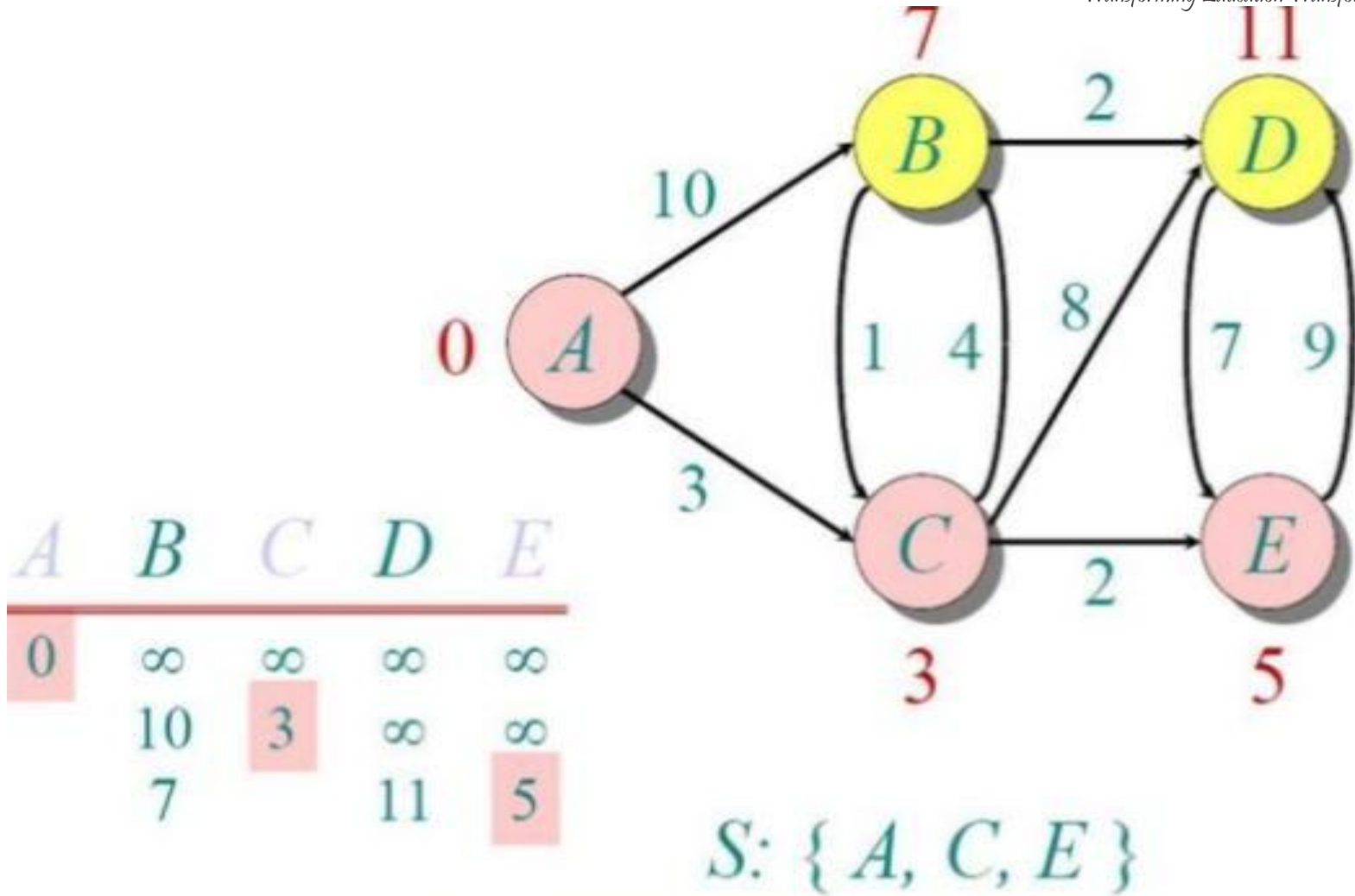


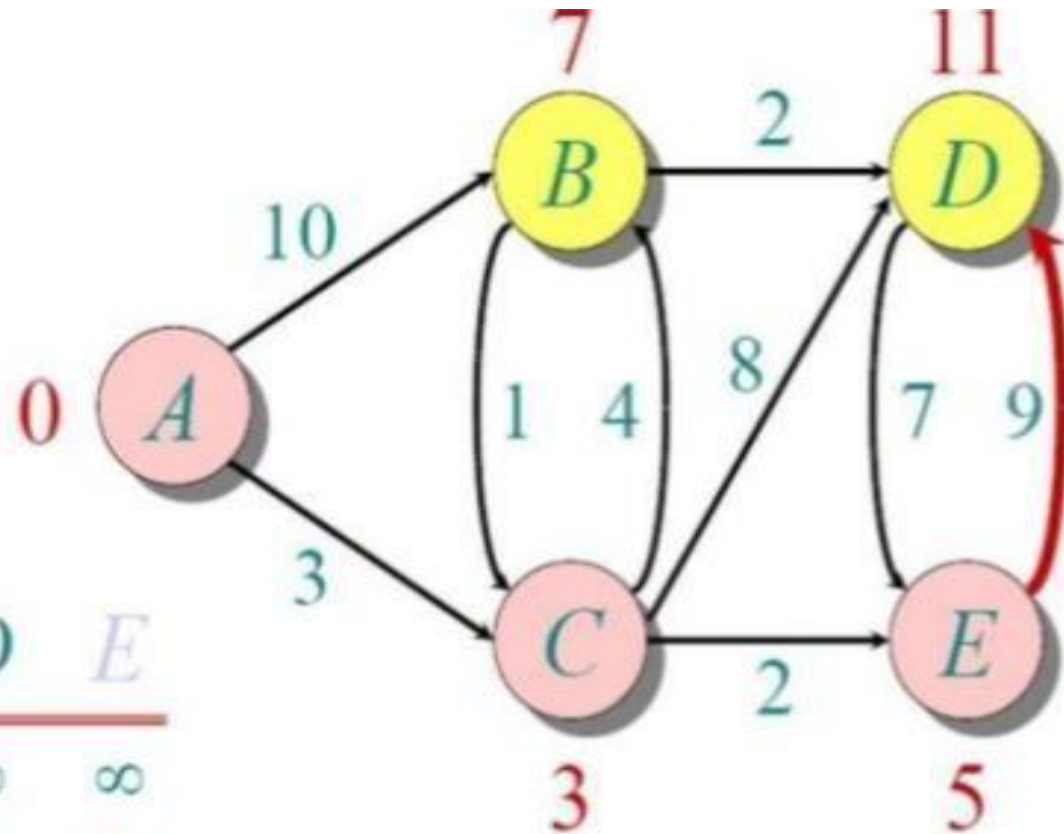
$S: \{A, C\}$



| <i>A</i> | <i>B</i> | <i>C</i> | <i>D</i> | <i>E</i> |
|----------|----------|----------|----------|----------|
| 0        | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
|          | 10       | 3        | $\infty$ | $\infty$ |
|          | 7        |          | 11       | 5        |

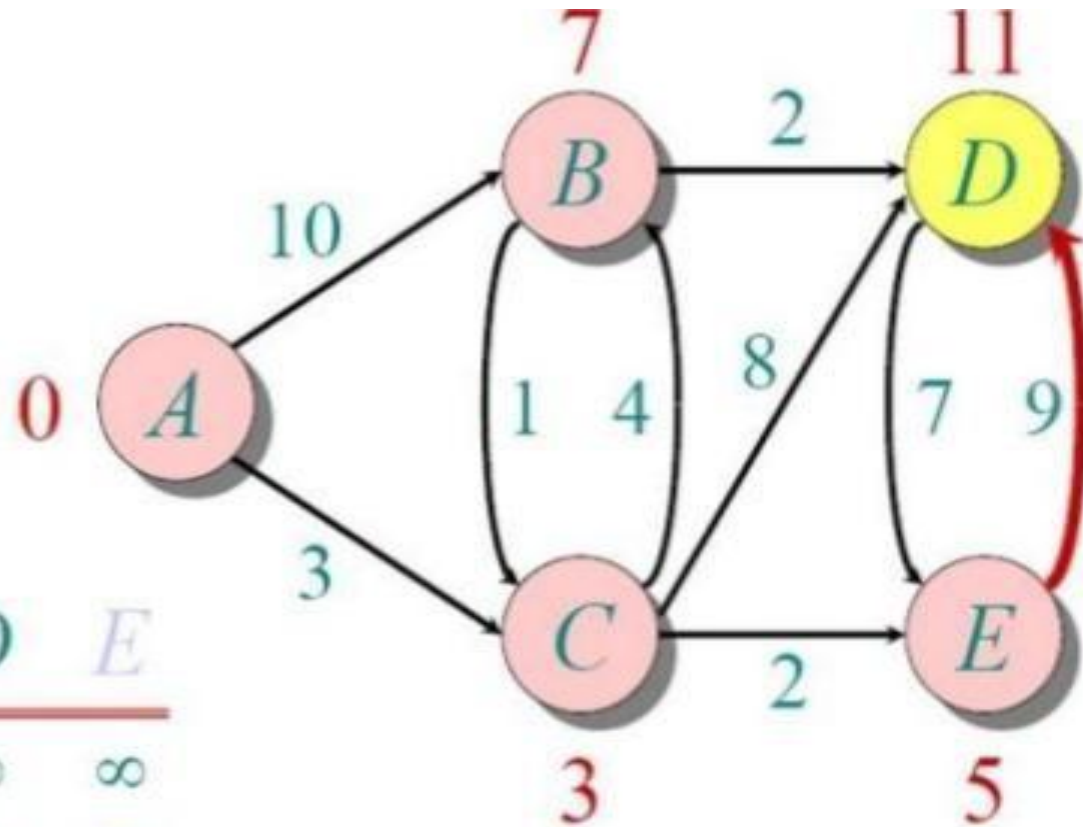
$S: \{A, C\}$





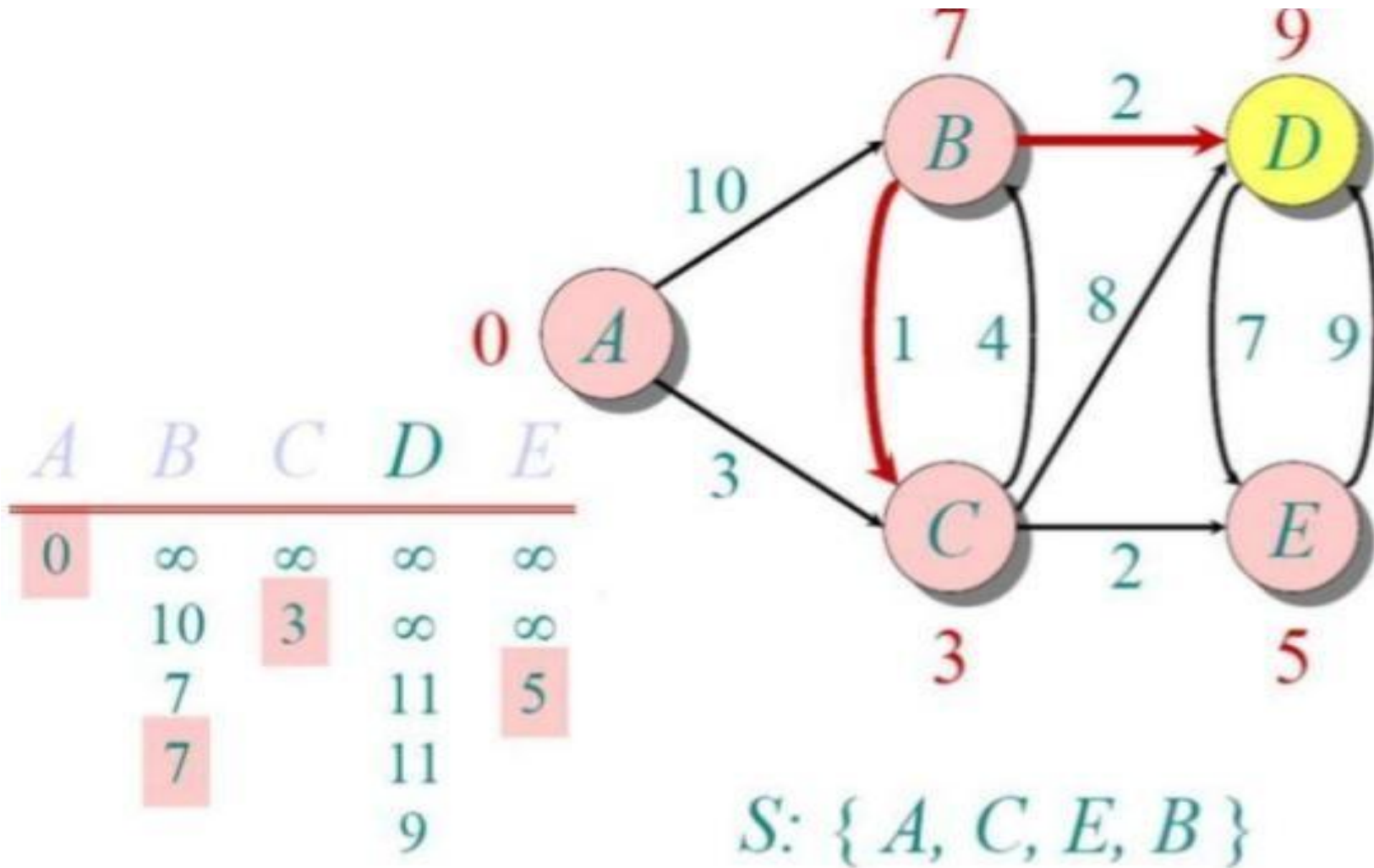
| A | B        | C        | D        | E        |
|---|----------|----------|----------|----------|
| 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
|   | 10       | 3        | $\infty$ | $\infty$ |
|   | 7        |          | 11       | 5        |
|   | 7        |          | 11       |          |

$S: \{A, C, E\}$



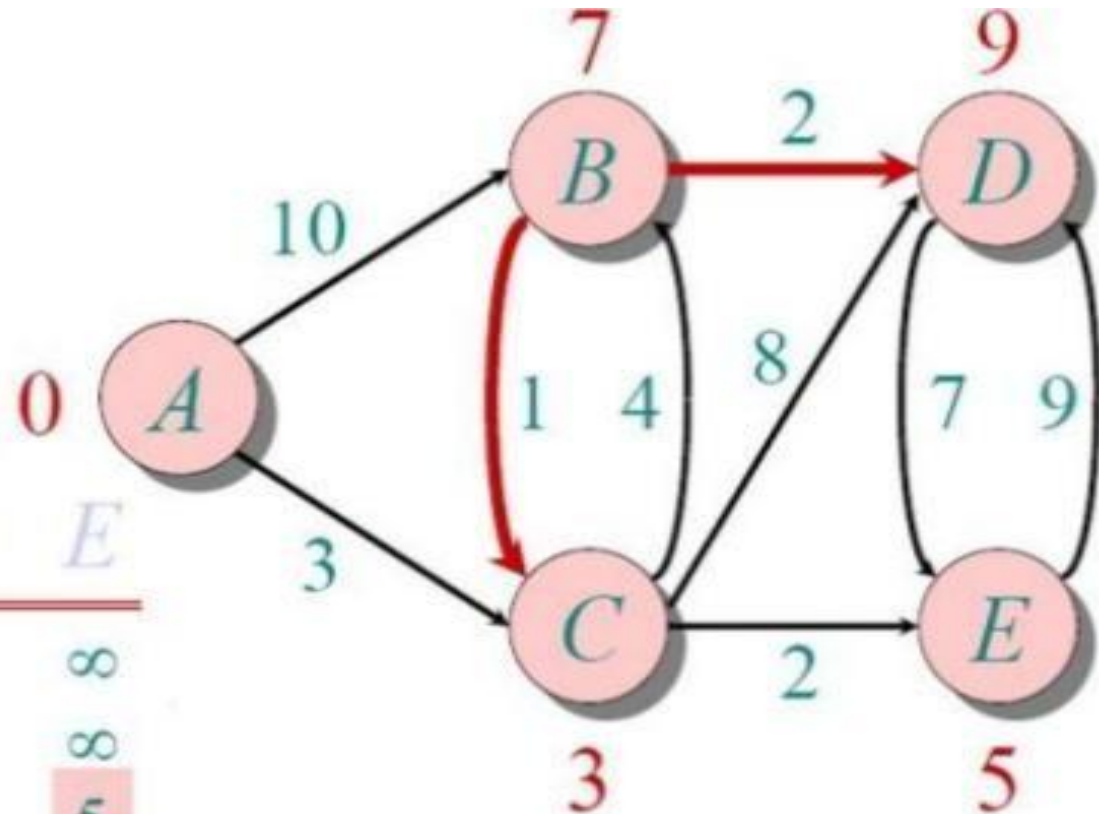
| <i>A</i> | <i>B</i> | <i>C</i> | <i>D</i> | <i>E</i> |
|----------|----------|----------|----------|----------|
| 0        | ∞        | ∞        | ∞        | ∞        |
|          | 10       | 3        | ∞        | ∞        |
|          | 7        |          | 11       | 5        |
|          | 7        |          | 11       |          |

$S: \{A, C, E, B\}$



A-C-B : 7  
 A-C : 3  
 A-C-B-D : 9  
 A-C-E : 5

| <i>A</i> | <i>B</i> | <i>C</i> | <i>D</i> | <i>E</i> |
|----------|----------|----------|----------|----------|
| 0        | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
|          | 10       | 3        | $\infty$ | $\infty$ |
|          | 7        |          | 11       | 5        |
|          | 7        |          | 11       |          |
|          |          |          | 9        |          |



$S: \{ A, C, E, B, D \}$

# Solution:



Shortest paths from s to other vertices

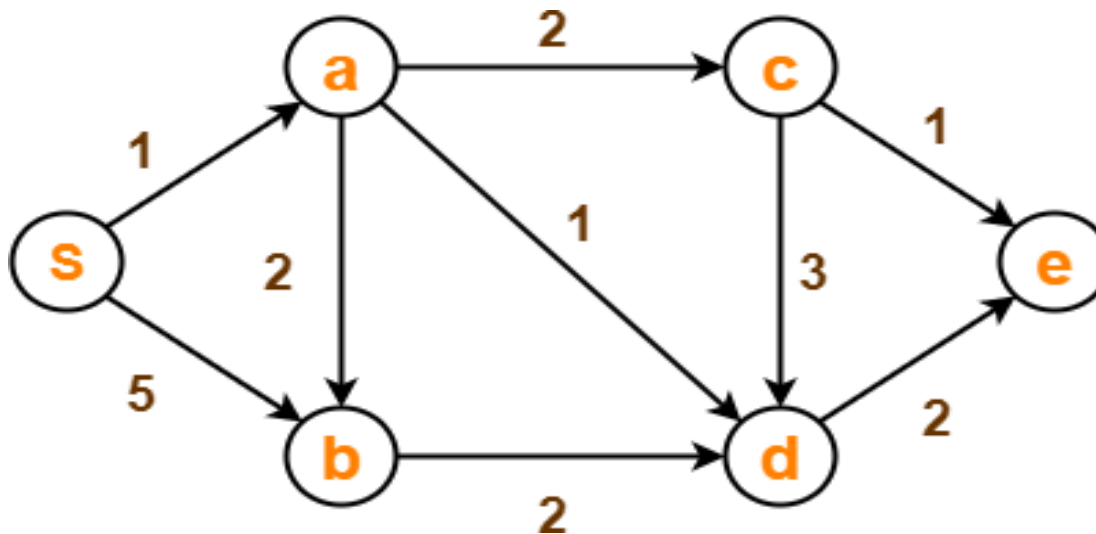
A-C-B 7 :

A-C: 3

A-C-B-D:9

A-C-E:5

Using Dijkstra's Algorithm, find the shortest distance from source vertex 'S' to remaining vertices in the following graph



# Solution

Shortest paths from s to other vertices

s-a :1

s-a-b:3

c. :3

d. :2

s-a-d-e:4

